Time-varying Minimum Variance Portfolio

Qingliang Fan[a], Ruike Wu[b], Yanrong Yang[c], and Wei Zhong[b]

[a]Department of Economics, The Chinese University of Hong Kong
[b]WISE & School of Economics, Xiamen University
[c]College of Business & Economics, The Australian National University

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Abstract

This paper proposes a new time-varying minimum variance portfolio (TV-MVP) in a large investment universe of assets. Our method extends the existing literature of minimum variance portfolio by allowing for time-varying factor loadings, which is the facilitator to capture the dynamics of asset returns' covariance structure (hence the optimal investment strategy in a dynamic setting). We also use a shrinkage estimation method based on a quasi-likelihood function to regularize the residual covariances further. We establish the desired theoretical properties of proposed time-varying covariance and the optimal portfolio estimators under a more realistic heavy-tailed distribution. Specifically, we provide consistency of the optimal Sharpe ratio of the TV-MVP and the sharp risk consistency. Moreover, we offer a test of constant covariance structure and show the asymptotic distribution of the test statistic. Simulations and empirical analysis suggest that the proposed TV-MVP has superior performance in estimation accuracy and out-of-sample Sharpe ratio, among other popular contemporary methods.

JEL Classification: G11, C38, C55, C58

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1 Introduction

1.1 MVP preliminaries

Minimum variance portfolio (MVP) optimization has a surging appearance in recent studies on portfolio management (Jagannathan and Ma, 2003; DeMiguel et al., 2009; Fan et al., 2012) pioneered by Markowitz's mean-variance model (Markowitz, 1952). Specifically, the classic MVP problem in a static setting is:

$$\begin{aligned} &\min\ W^\top \Sigma_r W, \\ &\text{s.t. } W^\top 1_p = 1. \end{aligned} \tag{1.1}$$

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where Σ_r is the constant p-dimensional population covariance matrix of asset returns; W is a $p \times 1$ vector of asset weights in the portfolio, and 1_p is a $p \times 1$ vector of 1's. The analytical solution for MVP weight is:

$$W^* = \frac{\Sigma_r^{-1} 1_p}{1_p^\top \Sigma_r^{-1} 1_p}. \tag{1.2}$$

The variance of MVP and the corresponding Sharpe ratio (SR) are:

$$R_{\min} = W^{*\top} \Sigma_r W^* = \frac{1}{1_p^\top \Sigma_r^{-1} 1_p}, \tag{1.3}$$

$$\mathrm{SR} = \frac{W^{*\top} \mu}{\sqrt{W^{*\top} \Sigma_r W^*}} = \frac{1_p^\top \Sigma_r^{-1} \mu}{\sqrt{1_p^\top \Sigma_r^{-1} 1_p}}. \tag{1.4}$$

respectively, and μ is a vector of expected returns of p stocks.

The sharp risk consistency (Ding et al., 2021) is achieved if

$$\frac{R(\widehat{W})}{R_{\min}} \xrightarrow{p} 1 \quad \text{as } T \to \infty. \tag{1.5}$$

where $R(\widehat{W}) = \widehat{W}^\top \widehat{\Sigma}_r \widehat{W}$ is the risk associated with the estimated portfolio, $\widehat{W}$, $\widehat{\Sigma}_r$ are the estimated portfolio weight and population covariance matrix, respectively. The literature has well documented that the true expected returns of assets are difficult to estimate, leading to the poor performances of mean-variance efficient portfolios in real applications (Jobson and Korkie, 1981; Michaud, 1989; Bai et al., 2009). The optimization problem (1.1) circumvents the estimation of expected returns and focuses solely on the risk (covariance matrix). The MVP is also shown to have a better out-of-sample Sharpe ratio compared to the mean-variance optimization portfolio and yield less extreme portfolio weights (Jorion, 1985; Best and Grauer, 1991; Jagannathan and Ma, 2003; DeMiguel et al., 2009).

1.2 The main challenges in modern investment practices and problem formation

This paper considers two noticeable problems of the classic MVP exposed by real applications in asset management. The first one is the high dimensionality of the portfolio due to the availability of assets in the modern investment universe. Investors often have to make decisions facing a portfolio with hundreds of stocks and other assets, the size (p) of which is often comparable or even larger than the sample periods (T) of the return data. It causes the "error maximization" issue of portfolio selection (Michaud, 1989). El Karoui (2010) characterized the statistical behavior of the empirical optimal vector and showed its bias in high-dimensionality. Ao et al. (2019) showed that when p/T is not too small, the plug-in mean-variance portfolios will fail to achieve risk constraints. The assumption of a constant, low-dimensional covariance matrix in (1.2) is not suitable in large portfolio construction.

Second, the turbulent financial market makes it difficult to pin down the optimal portfolio in a static manner. Unlike existing literature of minimum variance portfolio estimators under statistical factor models (Fan et al., 2013; Ding et al., 2021; Lee and Seregina, 2020), we consider the dynamics of the optimal portfolio and study the time-varying population covariance matrix. Motivated by both

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the evolution of the underlying production function of firms (Hoberg and Phillips, 2016; Buera et al., 2021) and the structural changes of the investment environment (Buraschi et al., 2010; Herrendorf et al., 2021), we model the dynamics of the portfolio's covariance structure explicitly using the time-varying factor loadings in a factor model as in Wang et al. (2021). The factor loadings measure the correlations between assets returns and factors. The time-varyingness in the factor loadings represents individual asset's distinct reactions to inherent risks associated with economic transition, preference change, and technological progress. In applications, we usually work with return data spanning a relatively long period covering different business cycles.

In this paper, we propose a new portfolio construction method called time-varying minimum variance portfolio (TV-MVP), with potentially a very large number of assets under consideration. The empirical investigation finds that our time-dependent investment strategy works very well in terms of out-of-sample performance when both return and risk are considered. We use statistical learning methods involving covariance matrix estimation and hypothesis testing to justify and robustify our portfolio's performance. The critical point of our approach is the decomposition of the population covariance matrix into the factor and idiosyncratic components, which facilitates the estimation and testing procedures. Specifically, we would apply the local principal component method proposed by Su and Wang (2017) to estimate time-varying factor loadings and corresponding factors. For the covariance matrix of idiosyncratic error, we assume it is constant and sparse. Such an assumption is strongly supported by empirical evidence[1]. We propose an estimator based on an ℓ_1 penalized quasi-likelihood function, which is different from the thresholding residual sample covariance method in Wang et al. (2021). The consistencies of covariance matrix estimator of returns, sparse covariance matrix estimator of residuals, and optimal weight estimator are derived in a theoretical framework. Moreover, we offer a test of constant covariance structure (the overall null hypothesis), which guides practitioners on whether to employ the proposed TV-MVP. Our test is "doubly robust" in the sense that the overall null is checked sequentially: the first step is to test whether the principal part of the covariance matrix is time-varying; the second step is to test whether the residual part of the covariance matrix is time-varying. The overall null hypothesis of constancy is not rejected only if both steps do not reject. In the empirical study using S&P 500 constituent stocks from 1995 to 2019, we find that around 70% of the time, the null is rejected in the first step, but not in the second step in a series of 299 tests based on rolling windows of size $T = 100$.

1.3 A brief review of the literature

Our research connects to the literature on covariance learning, MVP, and factor models. The high dimensional population covariance Σ is of research interest in many aspects of data science (Cai et al., 2011; Bickel and Levina, 2008; Fan et al., 2008; Ledoit and Wolf, 2017; Fan et al., 2020). Engle et al. (2019) combined the dynamic conditional correlation (DCC) using GARCH for the time-series dimension and nonlinear shrinkage of Ledoit and Wolf (2017) for the cross-sectional dimension in large dynamic covariance estimation. Testing covariance structures is also popular in the high-dimensional

[1]It is noteworthy that the constant factor-adjusted error covariance structure is not that stringent in practice. We can estimate an error covariance matrix for each rolling window period. From the perspective of investment strategy, our portfolio has the same generality as in Wang et al. (2021).

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statistical literature (Zheng et al., 2019; Zhong et al., 2019; Pelger and Xiong, 2020). A comprehensive overview of the high-dimensional covariance matrices study is provided in Fan et al. (2020). In recent years a surging strand of literature focused on the large dimensional MVP (Fan et al., 2012; Ledoit and Wolf, 2017; Lee and Seregina, 2020). Callot et al. (2021) provided a nodewise regression estimator for the inverse covariance matrix using Lasso. Ding et al. (2021) proposed a high dimensional MVP estimator extending POET (Fan et al., 2013) to enhance the portfolio's performance. Cai et al. (2020) studied high-dimensional MVP estimation based on high-frequency data. Factor models are prevalent in asset pricing literature. Using weighted two-pass cross-sectional regressions, Gagliardini et al. (2016) analyzed the path of time-varying risk premia in large portfolios. Giglio and Xiu (2021) developed a method to estimate the risk premium of an observable factor robust to missing factors.

1.4 Our contributions

Our paper is the first to develop theories for the time-varying minimum variance portfolio systematically. Although some existing methods are utilized in our study, the original contribution of the article is that we show their respective roles played in the new TV-MVP, which has not yet been fully examined in any of those methods alone. In developing the new theories under the framework of TV-MVP, such as the sharp risk consistency, we also extend the previous methods by establishing superior theoretical results in terms of convergence rates and general applicability under heavy-tailed distributions. Specifically, our main contributions are 1), we propose a new time-varying minimum variance portfolio, which is less restrictive than existing methods (Ding et al., 2021; Caner et al., 2020) and could accommodate the potential structural changes of how each asset react to common factors. The portfolio is more fitting to the real financial world, as shown in the real data applications; 2), we provide a quasi-likelihood estimator for the covariance matrix of the error component, which plays an important role in deriving more accurate estimation for the precision matrix of the assets returns; 3), we show the proposed method enjoys dynamic covariance and optimal weights estimation consistency; 4), for the first time, we prove the sharp risk consistency as in Ding et al. (2021), under the time-varying factor model and heavy-tailed distribution. We also show the Sharpe ratio consistency for TV-MVP; 5), we propose a test that could detect the time-varying factor loadings.

This paper will enrich the toolbox for active fund managers. Our proposed portfolio is easy to compute in real applications. Using S&P 500 Index constituent stocks' daily returns, we show the TV-MVP has preferable out-of-sample performance and its robustness during different business cycles.

The rest of the paper is organized as follows. In Section 2, we present TV-MVP model. Section 3 gives a brief description of the estimation procedure. Section 4 provides the main theoretical results. Section 5 introduces the test for constant covariances. Section 6 illustrates the finite sample performance of the proposed estimator using Monte Carlo simulations. An empirical application using S&P 500 index component stock returns follows in Section 7. Section 8 concludes the paper. Computational details and mathematical proofs are given in the supplementary material. We also briefly discuss the computational issues.

Throughout the paper, we use $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ to denote the minimum and maximum eigenvalue of a symmetric matrix A, respectively. $\mathrm{tr}(A)$ and $\det(A)$ compute the trace and determinant of

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A, respectively. We define the ℓ_1 norm, the operator norm and Frobenius norm of any matrix B respectively as $||B||_1 = \sum_{i,j} |B_{ij}|$, $\|B\| = \lambda_{\max}^{1/2}(B'B)$ and $\|B\|_F = \mathrm{tr}^{1/2}(B'B)$. We also use another two norms with notations $\|B\|_{\max} = \max_{ij} |b_{ij}|$ for $B = (b_{ij})_{p\times p}$, and $\|B - \Sigma\|_\Sigma = \frac{1}{\sqrt{p}} \left\|\Sigma^{-1/2} B \Sigma^{-1/2} - I_p\right\|_F$, where Σ is a given $p \times p$ positive definite matrix. $\Sigma_1 \succ \Sigma_2$ means $\Sigma_1 - \Sigma_2$ is positive definite. I_d is a d-dimensional identity matrix, $\mathbb{1}(\cdot)$ is an indicator function. We use $\xrightarrow{i.p.}$ and $\xrightarrow{i.d.}$ to denote convergence in probability and in distribution. Given a set $\mathcal{A}$, the cardinality of $\mathcal{A}$ is denoted by $|\mathcal{A}|$.

2 Time-varying minimum variance portfolio

First, the investment universe in this study is composed of p assets with observable return data. In the classic arbitrage pricing theory (Ross, 1976) and the approximate factor model of Chamberlain and Rothschild (1983), the expected return of financial assets is driven by several finite common factors. Specifically, the (excess) assets return can be modeled as:

$$r_{it} = b_{it}^\top F_t + e_{it}, \quad i = 1, 2, \ldots, p; \quad t = 1, 2, \ldots, T, \tag{2.1}$$

where r_{it} is the return of asset i at time t, F_t is the m-dimensional vector of common factors at time t, and b_{it} is the $m \times 1$ factor loading, which captures the relationship between common factors and financial asset i at time t. Due to the market turbulence, permanent shocks and a large number of assets in consideration, it is pretty stringent to assume that all the factor loadings b_{it}'s are constant all the time. e_{it} is the idiosyncratic error. In practice, factors can be either observed or not[2]. In this paper, we assume unobserved factors. In general, one can obtain the factors and loadings by principal component analysis.

The vector form of model (2.1) is

$$r_t = B_t F_t + e_t, \quad t = 1, 2, \ldots, T, \tag{2.2}$$

where $r_t = (r_{1t}, r_{2t}, \ldots, r_{pt})^\top$ is the vector of returns for p assets, $B_t = (b_{1t}, b_{2t}, \ldots, b_{pt})^\top$ is the factor loading matrix, and $e_t = (e_{1t}, e_{2t}, \ldots, e_{pt})^\top$.

Next, we turn to the covariance matrix of the assets, which is essential for portfolio weights. Imposing a factor structure (2.2) into financial return data helps us to decompose its covariance naturally:

$$\Sigma_{r,t} = B_t \Sigma_F B_t^\top + \Sigma_e, \quad t = 1, 2, \ldots, T, \tag{2.3}$$

where Σ_F denotes the covariance matrix of common factors, which is assumed to be time-invariant[3] and positive definite, and Σ_e is the covariance matrix of idiosyncratic errors (residuals). In the light of

[2]Some of those factors are commonly used by practitioners (Fama and French, 1992, 1993, 2015), while others might not be universally accepted. Harvey and Liu (2021), Feng et al. (2020) have good discussions on this 'zoo of factors'.

[3]A simple transformation of the factor model can guarantee the time-invariant of Σ_F, if the original Σ_F is time-varying. For instance, we can define new factors and factor loadings as $\widetilde{F}_t = \Sigma_F^{-1/2} F_t$ and $\widetilde{B}_t = B_t \Sigma_F^{1/2}$, respectively. Then $\widetilde{B}_t \widetilde{F}_t = B_t F_t$ and the covariance matrix of $\widetilde{F}_t$ is an identity matrix.

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(2.3), the time-varyingness of $\Sigma_{r,t}$ is due to the time-varying factor loadings B_t. The decomposition of the population covariance in (2.3) is crucial for our proposed time-varying optimal portfolio[4]. It paves the way for estimating and testing the population covariance matrix.

In large dimensional cases, the factor part of (2.3) has a low rank that can be efficiently estimated. For the residual part, it is common to use conditional sparsity assumption for Σ_e to regulate the model complexity (Ait-Sahalia and Xiu, 2017; Fan et al., 2020). Our estimation has the following key steps: 1. simultaneously estimate time-varying factor loading B_t and common factors F_t; 2. get the idiosyncratic residual, and further derive sparse covariance matrix estimator of residuals Σ_e; 3. estimate the covariance matrix of return by $\widehat{B}_t\widehat{\Sigma}_F\widehat{B}_t^\top + \widehat{\Sigma}_e$.

At last, the proposed TV-MVP strategy is to allocate assets using the following weights:

$$\widehat{W}_t = \frac{\widehat{\Sigma}_{r,t}^{-1}1_p}{1_p^\top\widehat{\Sigma}_{r,t}^{-1}1_p}. \tag{2.4}$$

3 The factor-augmented penalized likelihood estimation approach

The penalized quasi-likelihood[5] estimation approach is used to simultaneously estimate the factor loading B_t, the common factor F_t and the covariance matrix Σ_e of the residuals.

$$\left(\widehat{B}_t, \widehat{F}_t, \widehat{\Sigma}_e\right) = \arg\min_{(B_t,F_t,\Sigma_e)} \left\{\log\det\left(\Sigma_e\right) + tr\left(\Sigma_e^{-1}\widetilde{S}_e\right) + \|\Lambda * \Sigma_e\|_1\right\}, \tag{3.1}$$

where $\widetilde{S}_e = \frac{1}{T}\sum_{t=1}^T e_t e_t^\top$ with $e_t = r_t - B_t F_t$. Moreover, Λ is the non-negative arbitrary $p \times p$ tuning matrix, $*$ denotes the element-wise multiplication operator[6]. Note that we only penalize the off-diagonal elements of the covariance matrix. Thus the diagonal values of Λ are set to zeros.

The essential challenge of solving (3.1) comes from unobserved e_t due to unknown parameters B_t and oftentimes, unobservable F_t. To obtain a practical estimation for (3.1), we propose the following two-step estimation procedure: first, we estimate the factor model part (including factor loadings and common factors); second, with estimated error components, we solve the optimization problem (3.1).

3.1 Estimation of time-varying factor model

As discussed earlier, when returns are driven by underlying mechanisms, including technological progress and sentimental changes, the single-regime factor loading assumption over a long time period in the factor model is often too restrictive. We hence relax this assumption and consider factor loadings to be time-variant. Recall the time-varying factor model from Section 2:

$$r_t = B_t F_t + e_t, \quad t = 1, 2, \ldots, T.$$

[4]Although the precision matrix $\Sigma_{r,t}^{-1}$ is directly used in the portfolio computation, our interests focus on the covariance matrix due to that the covariance structure and its decomposition are integral parts in our study.

[5]The functional form of the non-penalized part of (3.1) is based on Gaussian errors $\{e_t\}_{t=1}^T$ in (2.2). Our theory is not dependent on this Gaussianity.

[6]We could conduct element-wise or uniform penalties on the elements of Σ_e, depending on the input Λ, which we elaborate on in the appendix.

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Let R denotes the $p \times T$ data matrix of assets' returns, and r_{it} is the (i,t)-th element of matrix R.

To simultaneously estimate time-varying factor loading and common factors, we can consider the local weighted least squares method which was proposed by Su and Wang (2017). Suppose the factor loading at time x around t, $b_{it} \approx b_{ix}$. Then we can solve the following optimization problem:

$$\min_{\{b_{ix}\}_{i=1}^{p},\{F_t\}_{t=1}^{T}} (pT)^{-1} \sum_{i=1}^{p} \sum_{t=1}^{T} (r_{it} - b_{ix}^{\top} F_t)^2 K_h \left(\frac{t-x}{T} \right), \tag{3.2}$$

where $K_h(\cdot) = h^{-1} K(\cdot/h)$, and $K(\cdot)$ is a kernel function and h is a bandwidth parameter. Equivalently, we can rewrite the above local weighted least squares problem (3.2) as the following:

$$\min_{\left(F^{(x)},B_x\right)} \operatorname{tr} \left(\left(R^{(x)} - F^{(x)} B_x^{\top} \right) \left(R^{(x)} - F^{(x)} B_x^{\top} \right)^{\top} \right), \tag{3.3}$$

where $R^{(x)} = \left(R_1^{(x)}, ..., R_p^{(x)} \right)$, $R_i^{(x)} = \left(k_{h,1x}^{1/2} r_{i1}, ..., k_{h,Tx}^{1/2} r_{iT} \right)^{\top}$ and $k_{h,tx} = h^{-1} K((t-x)/Th)$, for $i = 1, 2, \ldots, p$. $F^{(x)}$ is a $T \times m$ factor matrix, m is the number of factors, and B_x is the factor loading for time x. The identification conditions for (3.3) are:

$$F^{(x)^{\top}} F^{(x)}/T = I_m, \ B_x^{\top} B_x \text{ is a diagonal matrix.}$$

Following Su and Wang (2017), we know $\widehat{F}^{(x)}$ is $\sqrt{T}$ times m largest eigenvectors[7] of matrix $R^{(x)} R^{(x)^{\top}}$, and $\widehat{B}_x^{\top} = \widehat{F}^{(x)^{\top}} R^{(x)}/T$. Note that the estimator $\widehat{F}^{(x)}$ is consistent with a rotated factor. Therefore, the estimator for factor is given by $\widehat{F}_t = (\sum_{i=1}^{p} \widehat{b}_{it} \widehat{b}_{it}^{\top})^{-1} (\sum_{i=1}^{p} \widehat{b}_{it} r_{it})$ for $t = 1, 2, ..., T$ where $\widehat{B}_x = (\widehat{b}_{1x}, ..., \widehat{b}_{px})^{\top}$. Denote $\widehat{F}$ as the $m \times T$ matrix of estimated factors. It is easy to see the sample covariance matrix is $\widehat{\Sigma}_F = \frac{1}{T} \widehat{F} \widehat{F}^{\top}$. The number of factors can be determined using an information criterion detailed in the appendix.

Remark 1. Since a local nonparametric tool is referred, the boundary bias problem needs to be handled. We follow the method of Hong and Li (2005), Su and Wang (2017) to use the boundary kernel to solve it, that is

$$k_{h,t\tau}^{*} = h^{-1} K_{\tau}^{*} \left(\frac{t-\tau}{Th} \right) = \begin{cases} h^{-1} K\left(\frac{t-\tau}{Th}\right) / \int_{-(\tau/Th)}^{\infty} K(u) du, & \text{if } \tau \in [0, \lfloor Th \rfloor) \\ h^{-1} K\left(\frac{t-\tau}{Th}\right), & \text{if } \tau \in [\lfloor Th \rfloor, T - \lfloor Th \rfloor] \\ h^{-1} K\left(\frac{t-\tau}{Th}\right) / \int_{-\infty}^{(1-\tau/T)/h} K(u) du, & \text{if } \tau \in (T - \lfloor Th \rfloor, T] \end{cases}$$

where we take Epanechnikov kernel $K(u) = \frac{3}{4}(1-u^2) \mathbb{1}(|u| \leq 1)$. In fact, the choice of kernel function

[7] For the traditional principal component method, we need to solve

$$\min_{\{b_i\}_{i=1}^{p},\{F_t\}_{t=1}^{T}} \sum_{i=1}^{p} \sum_{t=1}^{T} (r_{it} - b_i^{\top} F_t)^2,$$

$$\text{s.t.} \ \ \frac{1}{T} F F^{\top} = I_m, \ B^{\top} B \text{ is diagonal.}$$

where F is $m \times T$ factor matrix, B is $p \times m$ loading matrix and R is $p \times T$ return matrix. And we can get $\widetilde{F} = \sqrt{T} eig_m(R^{\top} R)$ and $\widetilde{B} = \frac{1}{T} R \widetilde{F}^{\top}$, where $eig_m(A)$ takes the m largest eigenvectors of matrix A.

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has little influence to the final results. For the value of bandwidth, the Silverman's rule of thumb can be used, that is $h = (2.35/\sqrt{12})T^{-0.2}p^{-0.1}$. More discussions on the data-driven bandwidth choice is put in the appendix.

Remark 2. Before applying the time-varying factor model, it is better to test the existence of the time-varying factor loading structure. In this paper, we propose a sequential test of the constant covariance matrix that is detailed in Section 5.

3.2 Plug-in penalized likelihood estimation of the residual covariance matrix

Now we turn to the residual covariance matrix Σ_e. As discussed earlier, it is common to assume that the factor-adjusted residual covariance matrix is sparse, especially in large dimensions. We consider a sparse covariance matrix estimator for residuals to regulate the model complexity of Σ_e. Specifically, we impose the ℓ_1 penalty into the quasi-likelihood function and solve

$$\widehat{\Sigma}_e = \underset{\Sigma_e \succ 0}{\arg\min} \left\{ \log\det(\Sigma_e) + \operatorname{tr}\left({\Sigma_e}^{-1} S_e\right) + \|\Lambda * \Sigma_e\|_1 \right\}. \tag{3.4}$$

$S_e = \frac{1}{T}\sum_{i=1}^{T}(r_t - \widehat{B}_t\widehat{F}_t)(r_t - \widehat{B}_t\widehat{F}_t)^\top$ is the residual sample covariance matrix.

Solving Equation (3.4) seems to be a formidable challenge at first glance since the objective function is non-convex ($\log\det(\Sigma_e)$ is concave in Σ_e, and $\operatorname{tr}({\Sigma_e}^{-1} S_e)$ is convex in Σ_e). Here we solve this problem by applying a majorize-minimization algorithm. For simplicity, we consider Λ has common off-diagonal entries λ. We use a BIC approach similar to Fan et al. (2020) to select the tuning parameter λ. The detailed computation algorithm is put in the appendix.

4 Theoretical results

In this section, we establish the asymptotic properties for the time-varying covariance matrix $\widehat{\Sigma}_{r,t}$ and two criteria on portfolio evaluation: MVP Sharpe ratio and sharp risk consistency.

This section consists of three parts: the first part introduces regular assumptions and corresponding justifications; in the second part, we develop the rate of convergence for covariance matrices estimators $\widehat{\Sigma}_e$ and $\widehat{\Sigma}_{r,t}$; the third part provides the asymptotic evaluation for Sharpe ratio and sharp risk estimation.

4.1 Assumptions

In this part, we impose some regular assumptions on the time-varying factor model (2.1), including the common factors F_t, the time-varying factor loadings B_t and the error components e_t.

First, we consider the population covariance matrix $\Sigma_e = (\sigma_{ij})_{p\times p}$. As sparse structure is necessary on this matrix, we introduce some notations as follows. Let $\mathcal{S} = \{(i,j) : \sigma_{ij} \neq 0\}$, which is the set of nonzero elements in the matrix Σ_e. Denote $|\mathcal{S}|$ as the total number of nonzero elements in the matrix Σ_e. Then the number of nonzero off-diagonal elements is $s_1 := |\mathcal{S}| - p$. Furthermore, we define the row sparse cardinality $d = \max_{j=1,2,\ldots,p} |\mathcal{A}_j|$, where $\mathcal{A}_j = \{i : \sigma_{ji} \neq 0, i = 1, 2, \ldots, p\}$.

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Assumption 1. *There exist positive constants τ_1 and τ_2 such that*

$$0 < \tau_1 < \lambda_{\min}(\Sigma_e) \leq \lambda_{\max}(\Sigma_e) < \tau_2 < \infty. \tag{4.1}$$

Remark 1. *This assumption bounds extreme eigenvalues of the matrix Σ_e. The lower bound $\tau_1 > 0$ guarantees a well-defined precision matrix Σ_e^{-1}, which appears in optimal weights for the minimum variance portfolio.*

Secondly, we suppose that the common factor $\{F_t : t = 1, 2, \ldots, \}$ is an α-mixing process with the mixing coefficient $\alpha(n) \to 0$ as $n \to \infty$, where

$$\alpha(n) = \sup_{A \in \mathcal{F}_{-\infty}^{0}, B \in \mathcal{F}_{n}^{+\infty}} |P(A)P(B) - P(A \cap B)|,$$

with $\mathcal{F}_s^t$ denoting a σ-algebra generated by $\{F_\tau : s \leq \tau \leq t\}$. The α-mixing dependence condition is mild for a stationary and weakly dependent process. It is satisfied by commonly-used time series models such as the ARMA process. We impose the following conditions related to time-serial dependence in the model.

Assumption 2. *(a). $\{F_t : t = 1, 2, \ldots, \}$ is weak stationary and α-mixing process. The α-mixing coefficient decays to zero at a geometric rate, i.e. $\alpha(n) \leq C_0 \gamma^n$ with $0 < \gamma < 1$ and C_0 being a positive constant.*

(b). The error component $\{e_t : t = 1, 2, \ldots, T\}$ are independent and identically distributed (i.i.d) across the time $t = 1, 2, \ldots, T$.

Remark 2. *Assumption 2 assumes common factors are weak stationary time series while the error component is independent across the time period. Equivalently, common factors capture all time-serial dependence in the original data - asset returns. It is not an unusual assumption in literature, for instance, Lam and Yao (2012).*

The following assumption imposes moment conditions on the factors and error components.

Assumption 3. *There exist constants $C_1 > 0$ and $\theta_1 > 0$ such that*

$$\max_{1 \leq i \leq p} \mathbb{E}\left(\exp(\theta e_{it}^2)\right) \leq C_1, \quad \forall \theta \in [0, \theta_1], \tag{4.2}$$

$$\max_{1 \leq j \leq m} \mathbb{E}\left(\exp\left(\theta F_{jt}^2\right)\right) \leq C_1, \quad \forall \theta \in [0, \theta_1]. \tag{4.3}$$

Remark 3. *The moment conditions in (4.2) and (4.3) require exponential-type tails for the distribution of original data. It can be relaxed to the case of polynomial-type tails, which is more appropriate for financial applications. Details of discussions on this aspect are in the appendix.*

Assumption 4. *The kernel function $K(\cdot)$ is symmetric and continuously differential pdf function with compact support. As $p, T \to \infty$, the bandwidth h satisfies the following conditions*

$$h \to 0, \quad h \asymp T^{-\theta_2}, \quad \frac{T^{1-2\zeta} h}{\log^3(p \vee T)} \to \infty, \quad pT \exp(-\theta T^{\zeta}) = o(1), \tag{4.4}$$

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where θ_2 and ζ are two positive constants, and $0 < \theta < \theta_1$ with θ_1 being defined in Assumption 3.

Remark 4. *This assumption imposes common conditions on kernel functions and the bandwidth h. Here we impose the same assumptions on the bandwidth h, as Assumption 4 of Wang et al. (2021).*

Assumption 5. *(a). There exists a positive definite matrix $\Sigma_{b,t} \succ 0$ such that*

$$\sup_{1\le t\le T}\left\|\frac{1}{p}\sum_{i=1}^{p} b_{it}b_{it}^{\top} - \Sigma_{b,t}\right\| = o_p(1). \tag{4.5}$$

(b). The kernel weighted common factor $F_t^{(r)} := F_t K_{t,h}^{1/2}$ satisfies that

$$\sup_{1\le t\le T}\left\|\frac{1}{T}\sum_{t=1}^{T} F_t^{(r)}F_t^{(r)\top} - \Sigma_F\right\| = O_p\left(\sqrt{\frac{\log(T)}{Th}} + h^2\right), \tag{4.6}$$

where $\Sigma_F = \mathbf{E}\left(F_t F_t^{\top}\right)$.

Remark 5. *As analyzed in Wang et al. (2021), Assumption 5(a) implies that the minimum eigenvalue of $B_t B_t^{\top}$ is of the order p, which indicates that all factors are pervasive. The rate of convergence in Assumption 5(b) is also regular, as it can be derived under the α-mixing assumption imposed in Assumption 2. Please see Assumption 2 of Wang et al. (2021).*

It is worth noting that the estimation of the time-varying factor model is similar to that of Wang et al. (2021), while we utilize an alternative method - penalized quasi-likelihood estimation to approximate the precision matrix of the error component. In view of this point, assumptions on time-varying factor models (like Assumption 4 and Assumption 5) are similar to those in Wang et al. (2021).

Assumption 6. *Suppose the penalty parameter λ in penalized log-likelihood estimation (3.4) satisfies*

$$\omega(T,p,h) = o(\lambda), \tag{4.7}$$

where

$$\omega(T,p,h) = \sqrt{\frac{\log(T)}{ph} + \frac{\log(p\vee T)}{Th} + h^2}. \tag{4.8}$$

Moreover, the relationship between nonzero elements and the penalty parameter λ should satisfy

$$\min_{(i,j)\in\mathcal{S}} \frac{|\sigma_{ij}|}{\lambda} \to \infty, \quad as \quad n\to\infty. \tag{4.9}$$

Remark 6. *Assumption 6 provides two conditions on the penalty parameter λ. To obtain consistency of covariance matrix estimation, the order of $\omega(T,p,h)$ should be smaller than that of the penalty parameter λ. If $\min_{(i,j)\in S_1}|\sigma_{ij}|$ is a constant, (4.9) implies that $\lambda = o(1)$.*

The log-determinant function is of Bregman type because it is strictly convex, continuously differ-

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entiable and defined on the closed convex set $\mathcal{S}_+^p$, i.e.

$$\mathcal{S}_+^p := \{A \in R^{p\times p} : A = A^\top, A \succeq 0\}.$$

Here $\mathcal{S}_+^p$ consists of all $p \times p$ symmetric positive semi-definite matrices.

Consider the log-determinant function defined on $\mathcal{S}_+^p$,

$$g(A) := \begin{cases} -\log\det(A), & A \succ 0, \\ +\infty, & otherwise. \end{cases}$$

Denote the Hessian matrix of the log-determinant function (as a function of the covariance matrix Σ) as

$$\Gamma^* := \bigtriangledown_\Sigma^2 g(\Sigma)|_{\Sigma=\Sigma_e} = \Sigma_e^{-1} \otimes \Sigma_e^{-1}, \tag{4.10}$$

where $\otimes$ denotes the Kronecker matrix product and Σ_e is the true error term covariance matrix. So Γ^* is a $p^2 \times p^2$ matrix and its entry $\Gamma^*_{(i,j),(k,\ell)}$ is the second partial derivative $\frac{\partial^2 g}{\partial\Sigma_{ij}\partial\Sigma_{k\ell}}$ at the true value $\Sigma = \Sigma_e$.

Moreover, define the ℓ_∞-operator norm of the true covariance matrix $\Sigma_e = (\sigma_{ij}^e)_{i,j=1}^p$ as

$$\kappa_{\Sigma_e} := |||\Sigma_e|||_\infty = \max_{i=1,\ldots,p} \sum_{j=1}^p |\sigma_{ij}^e|.$$

Consider the sub-matrix of Γ^* below

$$\Gamma^*_{\mathcal{S}\mathcal{S}} = \left[\Sigma_e^{-1} \otimes \Sigma_e^{-1}\right]_{\mathcal{S}\mathcal{S}} \in R^{(s_1+p)\times(s_1+p)}.$$

Meanwhile, we denote the parameter

$$\kappa_\Gamma := \left|\left|\left|\Gamma^{*-1}_{\mathcal{S}\mathcal{S}}\right|\right|\right|_\infty.$$

Assumption 7. *There exists a constant $\alpha \in (0, 1]$ such that*

$$\max_{\varepsilon\in\mathcal{S}^c} \left|\left|\Gamma^*_{\varepsilon\mathcal{S}}\Gamma^{*-1}_{\mathcal{S}\mathcal{S}}\right|\right|_1 \le 1 - \alpha. \tag{4.11}$$

Remark 7. *This assumption limits the relation between the terms indexed by $\mathcal{S}^c$ and those from $\mathcal{S}$. To illustrate this point, we define*

$$Y_{(j,k),t} := e_{jt}e_{kt} - \mathbb{E}(e_{jt}e_{kt}), \quad \forall j, k = 1, 2, \ldots, p. \tag{4.12}$$

Then $\Gamma^*_{(j,k),(\ell,m)} = \mathbb{E}\left[Y_{(j,k),t}Y_{(\ell,m),t}\right]$. Let $Y_\mathcal{S} := \{Y_{(j,k)}, (j, k) \in \mathcal{S}\}$. Then the condition (4.11) reduces to

$$\max_{\varepsilon\in\mathcal{S}^c} \left|\left|\mathbb{E}(e_\varepsilon e_\mathcal{S}^\top)\mathbb{E}(e_\mathcal{S} e_\mathcal{S}^\top)\right|\right|_1 \le 1 - \alpha.$$

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4.2 Asymptotic properties of covariance matrices and precision matrices

In this part, we will study the rates of convergences for the quasi-likelihood estimator $\widehat{\Sigma}_e$ and the corresponding precision matrix $\widehat{\Sigma}_e^{-1}$. Based on these results, we could then obtain the rates of convergences for the time-varying covariance estimator $\widehat{\Sigma}_{r,t}$ and its inverse $\widehat{\Sigma}_{r,t}^{-1}$.

For the factor-augmented quasi-likelihood estimator $\widehat{\Sigma}_e$, we provide two styles of proofs in Theorem 1 and Theorem 2, respectively, to obtain rates of convergence under various norms.

Theorem 1. *Under Assumptions 1-6, there exists a local minimizer $\widehat{\Sigma}_e$ to the objective function (3.4) and*

$$\left\|\widehat{\Sigma}_e - \Sigma_e\right\|_F = O_p\left(\sqrt{s_1+p}\cdot\omega(T,p,h)\right), \tag{4.13}$$

where $\omega(T,p,h)$ is defined in (4.8).

Further, if $\sqrt{s_1+p}\cdot\omega(T,p,h) = o(1)$, the plug-in estimator $\widehat{\Sigma}_e$ is invertible and its rate of convergence is

$$\left\|\widehat{\Sigma}_e^{-1} - \Sigma_e^{-1}\right\|_F = O_p\left(\sqrt{s_1+p}\cdot\omega(T,p,h)\right). \tag{4.14}$$

Remark 8. *The convergence rates in (4.13) and (4.14) are related to the sparsity (quantified by s_1) as well as the order of estimators for factor loadings and common factors (i.e., $\omega(T,p,h)$).*

In the following theorem, similar to Ravikumar et al. (2011), we adopt the primal-dual witness method to obtain the rate of convergence for the estimator $\widehat{\Sigma}_e$ under the ℓ_∞ norm. And then, the rates of convergence under the spectral norm and Frobenius norm are derived directly.

Theorem 2. *Under Assumptions 1-7, we have, there exists a local minimizer $\widehat{\Sigma}_e$ to the objective function (3.4) and*

$$\begin{aligned}
\left\|\widehat{\Sigma}_e - \Sigma_e\right\|_{\max} &= O_p\left(\omega(T,p,h)\right),\\
\left\|\widehat{\Sigma}_e - \Sigma_e\right\|_F &= O_p\left(\sqrt{s_1+p}\cdot\omega(T,p,h)\right),\\
\left\|\widehat{\Sigma}_e - \Sigma_e\right\| &= O_p\left(\min\{\sqrt{s_1+p},d\}\cdot\omega(T,p,h)\right).
\end{aligned}$$

If $\omega(T,p,h) = o(1)$, we have

$$\left\|\widehat{\Sigma}_e^{-1} - \Sigma_e^{-1}\right\|_{\max} = O_p\left(\omega(T,p,h)\right).$$

If $\sqrt{s_1+p}\cdot\omega(T,p,h) = o(1)$, we have

$$\left\|\widehat{\Sigma}_e^{-1} - \Sigma_e^{-1}\right\|_F = O_p\left(\sqrt{s_1+p}\cdot\omega(T,p,h)\right).$$

If $\min\{\sqrt{s_1+p},d\}\omega(T,p,h) = o(1)$, we have

$$\left\|\widehat{\Sigma}_e^{-1} - \Sigma_e^{-1}\right\| = O_p\left(\min\{\sqrt{s_1+p},d\}\cdot\omega(T,p,h)\right).$$

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Remark 9. *Recall that d is the maximal number of nonzero elements in each row of the matrix Σ_e. Under the spectral norm, the rate of convergence for $\widehat{\Sigma}_e$ is lower than that under the Frobenius norm, if d is of smaller order than $\sqrt{s_1+p}$.*

Due to the existence of strong factors in time-varying factor model, we consider the following new defined relative norm

$$\left\|\widehat{\Sigma}_{r,t}-\Sigma_{r,t}\right\|_{\Sigma_{r,t}} := \frac{1}{\sqrt{p}}\left\|\Sigma_{r,t}^{-1/2}\widehat{\Sigma}_{r,t}\Sigma_{r,t}^{-1/2}-I_p\right\|_F.$$

We provide the rate of convergence for the time-varying covariance matrix estimator $\widehat{\Sigma}_{r,t}$.

Theorem 3. *Under Assumptions 1-7, we have*

$$\sup_t\left\|\widehat{\Sigma}_{r,t}-\Sigma_{r,t}\right\|_{\max} = O_P\left(\omega(T,p,h)\right),$$
$$\sup_t\left\|\widehat{\Sigma}_{r,t}-\Sigma_{r,t}\right\|_{\Sigma_{r,t}} = O_P\left(\max\{\sqrt{s_1+p},d\}\cdot\omega(T,p,h)+\sqrt{p}\cdot\omega^2(T,p,h)\right).$$

Furthermore, if $\max\{\sqrt{s_1+p},d\}\omega(T,p,h)=o(1)$*, we have*

$$\sup_t\left\|\widehat{\Sigma}_{r,t}^{-1}-\Sigma_{r,t}^{-1}\right\| = O_P\left(\max\{\sqrt{s_1+p},d\}\cdot\omega(T,p,h)\right),$$

Remark 10. *For the precision covariance matrix estimation of asset returns, Theorem 3 demonstrates that its convergence order is the same as that of the error component under the spectral norm in Theorem 2.*

4.3 Asymptotic properties for sharp risk and Sharpe ratio

Now we consider rates of convergence for the Sharpe ratio estimator and minimum risk estimator, respectively.

Recall that, with the estimators $\left(\widehat{B}_t,\widehat{F}_t,\widehat{\Sigma}_e\right)$, the estimator for $\Sigma_{r,t}$ is

$$\widehat{\Sigma}_{r,t}=\widehat{B}_t\widehat{B}_t^\top+\widehat{\Sigma}_e.$$

The estimated time-varying weights are then

$$\widehat{W}_t=\frac{\widehat{\Sigma}_{r,t}^{-1}1_p}{1_p^\top\widehat{\Sigma}_{r,t}^{-1}1_p}.$$

The associated minimum risk is

$$R_t\left(\widehat{W}_t\right)=\widehat{W}_t\widehat{\Sigma}_{r,t}\widehat{W}_t. \tag{4.15}$$

With the true covariance matrix $\Sigma_{r,t}$, the minimum risk (also called "oracle risk") is

$$R_{\min,t}=\frac{1}{1_p^\top\Sigma_{r,t}^{-1}1_p}.$$

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Next, let us look at the Sharpe ratio estimation. Based on the model (2.1), the population mean of the asset return r_t can be estimated by

$$\widehat{\mu}_t = \widehat{B}_t \frac{1}{T} \sum_{t=1}^{T} \widehat{F}_t.$$

Then the Sharpe Ratio estimator is

$$\widehat{SR}_t = \frac{1_p^\top \widehat{\Sigma}_{r,t}^{-1} \widehat{\mu}_t}{\sqrt{1_p^\top \widehat{\Sigma}_{r,t}^{-1} 1_p}}, \tag{4.16}$$

where

$$SR_t = \frac{1_p^\top \Sigma_{r,t}^{-1} \mu_t}{\sqrt{1_p^\top \Sigma_{r,t}^{-1} 1_p}}.$$

Assumption 8. *The minimum risk* $\sup_t R_{\min,t} = \sup_t \frac{1}{1^\top \Sigma_{r,t}^{-1} 1_p} \asymp p^{1-\eta}$*, where* η *is a constant satisfying* $p^\eta \min\{\sqrt{s_1+p}, d\} \cdot \omega(T,p,h) = o(1)$.

Assumption 9. *Suppose the term* $\sup_t 1^\top \Sigma_{r,t}^{-1} \mu_t \asymp p^{1-\delta}$*, where* δ *is a constant satisfying* $p^\delta \min\{\sqrt{s_1+p}, d\} \cdot \omega(T,p,h) + p^\delta \sqrt{p}\omega(T,p,h) = o(1)$.

Now we compare the minimum risk associated with $\widehat{W}_t$ and the oracle risk via their ratio. With a similar ratio criterion, we evaluate the Sharpe ratio estimation.

Theorem 4. *In addition to Assumptions in Theorem 3, we suppose Assumption 8 is also satisfied. Then we have the convergence rate for minimum risk estimator in (4.15), i.e.*

$$\sup_t \left| \frac{R_t\left(\widehat{W}_t\right)}{R_{\min,t}} - 1 \right| = O_p\left(p^\eta \min\{\sqrt{s_1+p}, d\}\omega(T,p,h)\right).$$

Remark 11. *This result for minimum risk is consistent with that in Ding et al. (2021) when the factor loading is treated as time-invariant.*

Theorem 5. *In addition to Assumptions in Theorem 3, we suppose Assumption 9 is also satisfied. Then we have the convergence rate for Sharpe ratio estimator in (4.16), i.e.*

$$\sup_t \left| \frac{\widehat{SR}_t}{SR_t} - 1 \right| = O_p\left(p^\delta \min\{\sqrt{s_1+p}, d\}\omega(T,p,h) + p^\delta \sqrt{p}\omega(T,p,h)\right).$$

Remark 12. *The order of Sharpe ratio consists of two parts: the first part is from the estimation of the precision matrix* $\widehat{\Sigma}_{r,t}^{-1}$ *while the second part is from the estimation for the population* $\widehat{\mu}_t$.

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5 Testing for constancy of covariance matrix over time

In this section, we propose a formal two-step test for the constancy of the population covariance matrix over time. Test statistics are presented here and its finite sample performances are illustrated in the simulation part accordingly.

Generally, the population covariance matrix of r_t is

$$\Sigma_{r,t} = B_t \Sigma_F B_t^\top + \Sigma_{e,t}, \tag{5.1}$$

where $\Sigma_{e,t}$ is the population covariance matrix of the error component e_t.

From the model (5.1), the time-varying features of $\Sigma_{r,t}$ are incurred by two parts: the time-varying factor loading B_t and time-varying error covariance structure $\Sigma_{e,t}$.

Next, we first illustrate the formal setting of the hypothesis testing, and then propose two test statistics to detect time-varyingness for factor loadings and error covariance structure, respectively.

5.1 Hypothesis testing

In order to test the constancy of $\Sigma_{r,t}$ over time $t = 1, 2, \ldots, T$, we conduct the following two-step sequential hypothesis test.

Step 1 :

$$\mathbb{H}_0^{(1)} : B_t = B_0, \quad for \quad t = 1, 2, \ldots, T; \; versus \; \mathbb{H}_a^{(1)} : B_t \neq B_0, \quad for \quad some \; t. \tag{5.2}$$

Step 2 :

$$\mathbb{H}_0^{(2)} : \Sigma_{e,t} = \Sigma_e, \quad for \quad t = 1, 2, \ldots, T; \; versus \; \mathbb{H}_a^{(2)} : \Sigma_{e,t} \neq \Sigma_e, \quad for \quad some \; t, \tag{5.3}$$

where B_0 is an $m \times p$ deterministic factor loading matrix and Σ_e is a $p \times p$ time-invariant covariance matrix.

In principle, the order of the two hypothesis testing in the sequential test above could be changed. In other words, as the first step, we can either test the time-varyingness for factor loadings or error covariance structure.

Next, we propose the test statistics for these two steps, respectively.

5.2 Test statistics

For the hypothesis testing in Step 1, we utilize the test statistic proposed in Su and Wang (2017) to test the constancy of factor loadings. In detail, the test statistic is

$$\widehat{M} = \frac{1}{pT} \sum_{i=1}^{p} \sum_{t=1}^{T} (\widehat{b}_{it}^\top \widehat{F}_t - \widetilde{b}_{i0}^\top \widetilde{F}_t)^2, \tag{5.4}$$

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where $\widetilde{b}_{i0}$ and $\widetilde{F}_t$ are estimators of the factor loading and common factors derived in Bai and Ng (2002) when the data follows a static factor model, i.e

$$r_{it} = b_{i0}^\top F_t + e_t,$$

where the factor loading b_{i0} is time-invariant.

To make self-contained content, we present the normalized version of the test statistic in practice here.

$$\widehat{J}_{pT} = \widehat{\mathbb{V}}_{pT}^{-1/2} \left(T p^{1/2} h^{1/2} \widehat{M} - \widehat{\mathbb{B}}_{pT} \right), \tag{5.5}$$

where $\widehat{\mathbb{V}}_{pT}$ and $\widehat{\mathbb{B}}_{pT}$ are given by

$$\widehat{\mathbb{V}}_{pT} = 2T^{-2}p^{-1}h^{-1} \sum_{i \le s \ne r \le T} \bar{K}(\frac{s-r}{Th})^2 (\widehat{F}_s^\top \widehat{\Sigma}_f \widehat{F}_r)^2 (\widehat{e}_r^\top \widehat{e}_s)^2,$$

$$\widehat{\mathbb{B}}_{pT} = \frac{h^{1/2}}{T^2 p^{1/2}} \sum_{i=1}^{p} \sum_{t=1}^{T} \sum_{s=1}^{T} (k_{h,st} \widehat{F}_s^\top \widehat{F}_t - \widetilde{F}_s^\top \widetilde{F}_t)^2 \widehat{e_{is}}^2$$

$\widehat{e}_{it}$ is the residuals of time-varying factor model, $\bar{K}(u) = \int_{-1}^{1} K(v)K(u-v)dv$ is the two-fold convolution kernel of $K(\cdot)$. The asymptotic distribution of $\widehat{J}_{pT}$ is standard normal distribution, which means when $\widehat{J}_{pT} > 1.96$, we can reject the null hypothesis that time-invariant factor loading with 95% confidence level.

Theorem 4.1 of Su and Wang (2017) established the asymptotic distribution for the test statistic $\widehat{J}_{pT}$, which is a standard normal distribution under the null hypothesis $\mathbb{H}_0^{(1)}$.

For the hypothesis testing in Step 2, we propose a new test statistic based on the spectral properties of the sample covariance matrix for the error component $\widehat{e}_t$ under the null hypothesis $\mathbb{H}_0^{(2)}$.

With the estimation of factor loadings $\widehat{B}_t$, the estimator for error components is

$$\widehat{e}_t = r_t - \widehat{B}_t \widehat{F}_t$$

Then the sample covariance matrix of $\widehat{e}_t$ is

$$S_e = \frac{1}{T} \widehat{e}\widehat{e}^\top = \frac{1}{T} \sum_{t=1}^{T} \widehat{e}_t \widehat{e}_t^\top,$$

where $\widehat{e} = (\widehat{e}_1, \widehat{e}_2, \ldots, \widehat{e}_T)$.

Now we analyze this sample covariance matrix S_e under the null hypothesis $\mathbb{H}_0^{(2)}$. Suppose that the error component e_t follows a linear independent structure

$$e_t = \Sigma_e^{1/2} \varepsilon_t, \tag{5.6}$$

where ε_t consists of independent and identically distributed (i.i.d) elements. In terms of (5.6), the linear independent structure implies that each element of the error components could be represented

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by a linear combination of i.i.d random variables. Moreover, under (5.6), the sample covariance matrix S_e can be expressed in the following form

$$S_e = \frac{1}{T} M_{\widehat{B}_t} \Sigma_e^{1/2} \varepsilon \varepsilon^\top \Sigma_e^{1/2} M_{\widehat{B}_t} =: \frac{1}{T} \boldsymbol{T} \boldsymbol{Y} \boldsymbol{Y}^\top \boldsymbol{T}, \tag{5.7}$$

where $\varepsilon = (\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_T)$, $\boldsymbol{T} = M_{\widehat{B}_t} \Sigma_e^{1/2}$, $\boldsymbol{Y} = \varepsilon$.

Under such data structure (5.6) with time-invariant covariance Σ_e, the sample covariance matrix S_e has a spectral feature: the limit of the empirical spectral distribution (ESD) F^{S_e} is a deterministic and well-known distribution. Here the ESD is defined as

$$F^{S_e}(y) = \frac{1}{p} \sum_{i=1}^{p} \mathbb{1}(\mu_i \leq y).$$

The limiting of the ESD of the sample covariance matrix S_e (called LSD) is derived in Lemma 1 of Pan et al. (2014). Now we utilize the normalized linear spectral statistic proposed in Pan et al. (2014) to check the time-invariant property for the error covariance structure. Consider the linear spectral statistic of the sample covariance matrix S_e

$$M_p = \int f(\zeta) dG_p(\zeta),$$

where $G_p(\zeta) = p\left(F^{S_e}(\zeta) - F_{c_p,H_p}\right)$ and $f(\zeta)$ is an analytic function. Here $F_{c_p,H_p}(\zeta)$ is obtained from the LSD $F_{c,H}(\zeta)$ of the sample covariance matrix S_e under the null hypothesis $\mathbb{H}_0^{(2)}$. To make the content self-contained, we introduce the test statistic as follows, although the details are included in Pan et al. (2014). We choose two simple test functions $f_1(\zeta) = \zeta$ and $f_2(\zeta) = \zeta^2$ for simplicity and consider their linear combination. Then we divide T time observations into two groups, each containing $[T/2]$ time series, where $[T/2]$ is the largest integer not greater than $T/2$. When T is odd, we discard the last time series. By either Theorem 2 in Pan et al. (2014) we have

$$\Big(\int x dG_p^{(i)}(x), \int x^2 dG_p^{(i)}(x) \Big) \xrightarrow{d} \Big(X_{(i),x}, X_{(i),x^2} \Big), \ as\ p, T \to \infty, \ i = 1, 2, \tag{5.8}$$

where $G_p^{(i)}(x) = p\left(F^{S_e^{(i)}}(x) - F_{c_{p^{(i)}},H_{p^{(i)}}}(x)\right)$ with $c_{p^{(i)}} = p/[T/2], H_{p^{(i)}} = H_p$, $F_{c_{p^{(i)}},H_{p^{(i)}}}(x)$ is the analog of F_{c_p,H_p} but corresponding to $S_e^{(i)} = \frac{1}{[T/2]} \widehat{e}_{(i)} \widehat{e}_{(i)}^\top$ and $\widehat{e}_{(i)}$ consisting of the i-th group of the divided time series for $i = 1, 2$. Let $\widehat{e} = (\widehat{e}_{(1)}, \widehat{e}_{(2)})$ for case where T is even. Here $\Big(X_{(i),x}, X_{(i),x^2} \Big)$ is the limiting distribution corresponding to the i-th group time series. Since the statistics on the left hand side of (5.8) for the two groups of time series are independent under $\mathbb{H}_0^{(2)}$, we calculate the difference between the two statistics and obtain

$$\Big(\int x d\widetilde{G}_p(x), \int x^2 d\widetilde{G}_p(x) \Big) \xrightarrow{d} \Big(\widetilde{X}_x, \widetilde{X}_{x^2} \Big), \ as\ n \to \infty, \tag{5.9}$$

where

$$\widetilde{G}_p(x) = G_p^{(1)}(x) - G_p^{(2)}(x) = p\Big(F^{S_e^{(1)}}(x) - F^{S_e^{(2)}}(x) \Big), \tag{5.10}$$

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and $\widetilde{X}_x = X_{(1),x} - X_{(2),x}$, $\widetilde{X}_{x^2} = X_{(1),x^2} - X_{(2),x^2}$.

It follows from Theorem 2 in Pan et al. (2014) that $(\widetilde{X}_x, \widetilde{X}_{x^2})$ is bivariate normal with mean 0 and covariance matrix $\widetilde{\mathbf{\Omega}}$, where $\widetilde{\mathbf{\Omega}} = 2\mathbf{\Omega}$ and $\mathbf{\Omega} = (\omega_{gh})_{2\times 2}$ is the asymptotic covariance matrix of $(\int x dG_p^{(i)}(x), \int x^2 dG_p^{(i)}(x))$ given by

$$\omega_{gh} = -\frac{1}{\pi^2} \oint_{\mathcal{C}_1} \oint_{\mathcal{C}_2} \frac{f_g(z_1) f_h(z_2)}{(\underline{m}_\phi(z_1) - \underline{m}_\phi(z_2))^2} \frac{d}{dz_2} \underline{m}_\phi(z_2) \frac{d}{dz_1} \underline{m}_\phi(z_1) dz_1 dz_2, \tag{5.11}$$

where $f_g(z) = z$ and $f_h(z) = z^2$.

Note that (5.10) does not involve any unknown parameters. However, our experience suggests that $\widetilde{G}_n(x)$ may not converge in distribution. Therefore, we propose using the following testing statistic for $\mathbb{H}_0^{(2)}$:

$$L_p = \Big(\int x d\widetilde{G}_p(x), \int x^2 d\widetilde{G}_p(x) \Big) \widetilde{\mathbf{\Omega}}^{-1} \begin{pmatrix} \int x d\widetilde{G}_p(x) \\ \int x^2 d\widetilde{G}_p(x) \end{pmatrix}. \tag{5.12}$$

Then, the test statistic L_p converges in distribution to a chi-squared distribution with two degrees of freedom.

Remark 13. *Note that the matrix $\boldsymbol{T}$ in (5.7) depends on the estimated factor loading via the projection matrix $M_{\widehat{B}_t}$. For the case of $\widetilde{\boldsymbol{T}} = M_{B_t} \Sigma_e^{1/2}$, the asymptotic theory in Pan et al. (2014) can be applied directly. As $M_{\widehat{B}_t}$ is a projection matrix of low rank, by similar proofs to Theorem 2 of Gao et al. (2017), we could show that, under the case of $M_{\widehat{B}_t}$, the asymptotic distribution of linear spectral statistics is still normal distribution with the same asymptotic variance as that for the case of M_{B_t}. Fortunately, in the test statistic L_p in (5.12), we utilize the difference of two linear spectral statistics, which results in zero asymptotic mean in (5.9). So we have the same asymptotic distribution for the test statistic L_p as in Pan et al. (2014).*

6 Simulation

We conduct various simulation studies to evaluate the finite sample performance of the proposed estimators. In the following subsections, we first study the estimation procedure. And then, we focus on hypothesis testing.

6.1 Simulation for portfolio estimators

6.1.1 Model Set-up

We first consider simulation DGPs following the factor model defined in (2.1) with $m = 2$ common factors that are generated as:

$$f_{1,t} = \mu_{f1} + 0.6 f_{1,t-1} + u_{1,t},\ u_{1,t} \overset{iid}{\sim} N(0, 1 - 0.6^2)$$

$$f_{2,t} = \mu_{f2} + 0.3 f_{2,t-1} + u_{2,t},\ u_{2,t} \overset{iid}{\sim} N(0, 1 - 0.3^2)$$

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where $u_{i,t}$ and μ_{fi} for i = 1,2 are disturbance and intercept terms, respectively, and $f_{1,0} = f_{2,0} = 0$. The following four kinds of time-varying factor loadings are considered. Note that once the factor loadings are generated or computed (from the real data), we fix them over 500 replications.

Setting (1): Step function type factor loadings. The factor loadings $b_{it} = (b_{it}^{(1)}, b_{it}^{(2)})^T$ are set to be the step functions as follows:

$$b_{it}^{(1)} = \begin{cases} 2|z_i^{(1)}| & \text{for} \quad 1 \leq t \leq D \\ 2|z_i^{(1)}| + z_i^{(2)} & \text{for} \quad D+1 \leq t \leq 2D \\ 2|z_i^{(1)}| + z_i^{(2)} + z_i^{(3)} & \text{for} \quad 2D+1 \leq t \leq 3D \\ \vdots & \\ 2|z_i^{(1)}| + \sum_{j=2}^{k} z_i^{(j)} & \text{for} \quad (k-1)D+1 \leq t \leq kD \\ \vdots & \end{cases}$$

where $z_i^{(j)}$ for $j = 1, 2, \ldots, K$ are standard normal random variables, and D is the length of each step ($K = T/D$ steps total). $b_{it}^{(2)}$ has the similar form as $b_{it}^{(1)}$ by replacing $z_i^{(j)}$ by $|v_i^{(j)}|$ and $v_i^{(j)} \sim N(0,1)$ for $j = 1, 2, \ldots, K$. For this DGP, the values of parameter D is given by 25, 50 and 100 for $p = 50, 100, 200$, respectively.

Setting (2): Time-varying smooth function type factor loadings. For this setting,

$$b_{it}^{(1)} = (3 + z_i^{(1)})t/T + z_i^{(2)} sin(4\pi t/T) + z_i^{(3)}$$

where $z^{(j)}, j = 1, 2, 3$ are standard normal random variables. And,

$$b_{it}^{(2)} = (3 + z_i^{(1)})t/T + z_i^{(2)} sin(4\pi t/T) + (z_i^{(4)} t/T)^2 + z_i^{(3)}$$

where the extra square term is added to distinguish it with $b_{it}^{(1)}$ and also amplify the variations in factor loadings.

In the first two DGPs, the values of μ_{f1} and μ_{f2} for common factors are set to zero, and the error terms e_{it}'s are drawn from the i.i.d. multivariate normal distribution with identity matrix as its covariance matrix. We use the rule of thumb bandwidth choice, $2.35/\sqrt{12}T^{-0.2}p^{-0.1}$, which is the same as Su and Wang (2017). The final results of the evaluation statistics (the details are in the next subsection) are averaged over the time points T, $T + D$ and $T + 2D$, where $D = 25$, 50 and 100 for $p = 50, 100, 200$, respectively.

Settings (3)-(4): Factor loading mimicking the S&P 500 monthly and daily data. For these two DGPs, the factor loadings are calibrated from monthly and daily stock excess returns of the component stocks listed in the S&P 500 index, which is obtained from the Center for Research in Security Prices (CRSP). Stocks are randomly selected for different portfolio sizes. For monthly data, the time range is from 1980/01 to 2018/06. For daily data, the time range is from 2006/01/03 to 2009/12/31. The constant covariance hypothesis is rejected using the proposed tests in the previous section. We apply the local principal component method with the rule of thumb bandwidth to compute the time-varying factor loadings and common factor estimators. We then use those estimates as the population values in

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the simulations. More preciously, $b_{it}^{(1)}$ and $b_{it}^{(2)}$ are given by factor loadings estimators, μ_{f1} and μ_{f2} use the mean values of common factor estimators, which are generally nonzero vectors. Moreover, based on local PCA results, a sparse covariance estimator of idiosyncratic residual $\hat{\Sigma}_e$ can be obtained by applying our shrinkage estimator. We then generate the error term e_{it} from i.i.d. $N(0, \hat{\Sigma}_e)$.

We also take the average over three different time points to capture the time-varying features. For monthly data in DPG 3, we take averages among time points T, $T+25$ and $T+50$ for all portfolio sizes. For daily data in DGP 4, we use the same average time period settings as the first two DGPs.

To assess the performance of the proposed estimator, the evaluation statistics for portfolios and covariance matrices are:

(1) Risk error:

$$\text{Risk error} = \left|\widehat{W}_t^\top \widehat{\Sigma}_{r,t} \widehat{W}_t - W_t^\top \Sigma_{r,t} W_t\right| = \left|\frac{1}{1_p^\top \widehat{\Sigma}_{r,t}^{-1} 1_p} - \frac{1}{1_p^\top \Sigma_{r,t}^{-1} 1_p}\right| \tag{6.1}$$

where $\hat{W}_t$ and W_t are estimated and true optimal weights at time point t, respectively. $\Sigma_{r,t}$ and $\hat{\Sigma}_{r,t}$ are the true population covariance matrix and its estimate at time point t, respectively.

(2) Weight error: the ℓ_2 norm of the difference between the estimated and true optimal weights.

$$\text{Weight error} = \|\widehat{W}_t - W_t\|_2 = \sqrt{\sum_i \left(\widehat{W}_t - W_t\right)_i^2} \tag{6.2}$$

(3) Sharpe ratio error: the difference between the estimated and true portfolio Sharpe ratios.

$$\text{SR error} = \left|\frac{\widehat{W}_t^\top \widehat{\mu}_t}{\sqrt{\widehat{W}_t^\top \widehat{\Sigma}_{r,t} \widehat{W}_t}} - \frac{W_t^\top \mu_t}{\sqrt{W_t^\top \Sigma_{r,t} W_t}}\right| \tag{6.3}$$

For the first two DGPs, since the expected asset return is 0, it is not meaningful to compare the Sharpe ratios. And for the last two DGPs, an estimator of the expected return is required to calculate values of the Sharpe ratio. Under our factor structure, we have the estimated expected return $\hat{\mu}_t = \hat{B}_t \frac{1}{T} \sum_{t=1}^{T} \hat{f}_t$, and for other comparing strategies, we use the sample mean.

(4) Covariance error: the Frobenius norm of the difference between estimated and true covariance of assets returns.

$$\text{Covariance error} = \|\Sigma_{r,t} - \hat{\Sigma}_{r,t}\|_F = \sqrt{\sum_{ij} \left(\Sigma_{r,t} - \hat{\Sigma}_{r,t}\right)_{ij}^2} \tag{6.4}$$

This metric reflects the goodness of the covariance matrix estimator of assets returns.

(5) Residual covariance error: the Frobenius norm of the difference between residual covariance estimator and population error covariance matrix.

$$\text{Residual Covariance error} = \|\Sigma_e - \hat{\Sigma}_e\|_F = \sqrt{\sum_{ij} \left(\Sigma_e - \hat{\Sigma}_e\right)_{ij}^2} \tag{6.5}$$

As stated in the covariance decomposition in Equation (2.3), the estimation of error covariance matrix

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Σ_e is essential for our model. For DGPs 1 and 2, the true $\Sigma_e = I$, for DGPs 3 and 4, the true Σ_e is not very sparse (around 50%), since it is estimated using the proposed method from the real data.

6.1.2 Other benchmark methods

We compare our TV-MVP with the following benchmark portfolios: (1) Sample: portfolio when the sample covariance matrix is applied. (2) Linear: portfolio using the linear shrinkage estimator proposed by Ledoit and Wolf (2003). (3) Nonlinear: portfolio when nonlinear shrinkage estimator proposed by Ledoit and Wolf (2017) is used. (4) Nodewise: portfolio when Node-wise sparse precision matrix estimator is applied (Callot et al., 2021). This approach estimates the precision matrix row by row based on the relationship between the precision matrix element and the ordinary least square estimator.

The following strategies (5), (6) and (7) are based on factor structures. (5) POET: portfolio when principal orthogonal complement thresholding approach proposed by Fan et al. (2013) is applied. (6) Fglasso: portfolio when factor graphical Lasso approach proposed by Lee and Seregina (2020) is applied. They use principal component analysis to compute time-invariant factor loadings, and then a weighted graphical Lasso for the residual precision matrix. (7) Fsparse: this portfolio is the constant factor loadings counterpart of our TV-MVP. It first uses PCA to extract common factors and then uses our sparse covariance matrix estimator.

(8) DCC-NL (Engle et al., 2019): Nonlinear shrinkage estimator of the DCC model (Engle, 2002). We use GARCH(1,1) for the dynamics of the univariate volatilities.

The simulation results from Tables 1 to 2 show the superiority of TV-MVP in all five error metrics in most cases. Under DGP 1 and DGP 2, TV-MVP shows robust performances with different patterns of time-varying factor loadings. “Fsparse”, which fails to consider the time-varying factor loadings, produces a much larger deviation than TV-MVP, in terms of covariance estimator, portfolio variance and optimal portfolio allocation. Related to our proposal is DCC-NL, which estimates the dynamic covariance matrix. It shows good performance in covariance error measures and beats TV-MVP in a few cases. And the overall performance of TV-MVP dominates DCC-NL.

The results under DGP 3 and DGP 4 are reported from Tables 3 to 4. Notably, the weight and Sharpe ratio errors of TV-MVP are the lowest in most cases. Furthermore, TV-MVP is also quite close to the best results for other error measures if it is not the best. For example, in Table 3 with $p = 100$, $T = 300$, the lowest covariance error 0.131 is obtained by DCC-NL, and our method is the second best at 0.144; in Table 4 with $p = 100$, $T = 300$, the lowest risk error belongs to “Nonlinear" with 7.78E-06, and TV-MVP based portfolio risk error is 9.56E-06. Notice that the results of TV-MVP with large p/T are sometimes unsatisfactory, e.g., in the monthly case with $p = 50$ and $T = 100$. Since the local nonparametric estimator of factor loadings uses the weighted local samples, it pays the price of requiring a relatively large sample size. When T is not large compared with portfolio size p and a small bandwidth is selected, some subpar estimations of factor loadings might exist during the whole period, especially at the boundary region.

Another point that needs attention is that the pattern of population factor loadings might affect the portfolio performance. See the left subfigure of Figure 1. Although considerable variation exists in

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Table 1: Simulation results based on DGP 1

	p = 50, T = 100				**T = 200**				**T = 300**			
	Risk	Weight	Cov	Σ_e	Risk	Weight	Cov	Σ_e	Risk	Weight	Cov	Σ_e
Sample	0.0938	0.5018	553.9	-	0.0224	0.3379	1914.1	-	0.0302	0.3908	2940.2	-
Linear	0.1349	0.2495	577.7	-	0.1830	0.2269	1938.9	-	0.4910	0.2351	2993.9	-
Nonlinear	0.0506	0.1839	560.8	-	0.0760	0.1885	1919.8	-	0.1016	0.2312	2952.5	-
Nodewise	0.1653	0.3901	1466.6	-	0.1687	0.3929	3519.7	-	0.2338	0.4839	7194.1	-
DCC-NL	0.0459	0.3638	**524.4**	-	0.0413	0.4260	1009.0	-	0.0877	0.5038	1962.4	-
POET	0.1051	0.1987	560.8	6.4	0.1384	0.2001	1920.3	12.6	0.3076	0.2390	2958.7	13.6
Fglasso	0.0753	0.4042	546.7	22.2	0.0211	0.3044	1909.2	37.7	0.0281	0.3489	2926.1	43.6
Fsparse	0.1102	0.1989	545.9	6.6	0.1417	0.2004	1907.9	12.8	0.3132	0.2395	2924.3	13.8
TV-MVP	**0.0229**	**0.1056**	556.6	**1.1**	**0.0084**	**0.0680**	**660.5**	**1.0**	**0.0155**	**0.0753**	**1711.8**	**1.1**
	p = 100, T = 200				**T = 300**				**T = 400**			
	Risk	Weight	Cov	Σ_e	Risk	Weight	Cov	Σ_e	Risk	Weight	Cov	Σ_e
Sample	0.0320	0.3215	1070.2	-	0.0246	0.2987	2239.5	-	0.0156	0.2773	3686.4	-
Linear	0.0185	0.2001	1095.1	-	0.0419	0.2133	2267.8	-	0.0645	0.2053	3717.6	-
Nonlinear	0.0198	0.1287	1077.4	-	0.0265	0.1553	2245.9	-	0.0312	0.1574	3692.3	-
Nodewise	0.0591	0.2410	3068.4	-	0.0920	0.2953	5044.1	-	0.1195	0.3325	7320.0	-
DCC-NL	0.0135	0.2762	**874.8**	-	0.0365	0.3459	**1329.01**	-	0.0344	0.3335	1892.5	-
POET	0.0395	0.1096	1078.7	10.7	0.0776	0.1466	2248.1	14.1	0.1025	0.1634	3694.2	17.5
Fglasso	0.0204	0.2408	1057.0	47.6	0.0119	0.2449	2223.4	58.9	0.0080	0.2311	3683.6	66.9
Fsparse	0.0406	0.1101	1056.8	10.9	0.0791	0.1470	2223.5	14.3	0.1039	0.1637	3682.8	17.7
TV-MVP	**0.0037**	**0.0432**	895.7	**1.1**	**0.0036**	**0.0438**	1441.2	**1.1**	**0.0103**	**0.0424**	**1729.5**	**1.4**
	p = 200, T = 500				**T = 600**				**T = 800**			
	Risk	Weight	Cov	Σ_e	Risk	Weight	Cov	Σ_e	Risk	Weight	Cov	Σ_e
Sample	0.0151	0.2174	3340.5	-	0.0121	0.2121	4600.2	-	0.0099	0.1906	7602.3	-
Linear	0.0029	0.1661	3369.3	-	0.0065	0.1701	4630.1	-	0.0087	0.1574	7631.7	-
Nonlinear	0.0107	0.0955	3347.3	-	0.0131	0.1099	4607.2	-	0.0111	0.0967	7608.1	-
Nodewise	0.0397	0.1921	8180.5	-	0.0460	0.2079	10279.9	-	0.0583	0.2336	14650.4	-
DCC-NL	0.0222	0.2325	**1788.0**	-	0.0275	0.2446	2035.2	-	0.0380	0.2596	2810.7	-
POET	0.0346	0.0888	3350.2	16.8	0.0370	0.0918	4609.5	20.0	0.0586	0.1116	7610.9	24.1
Fglasso	0.0128	0.1960	3319.3	100.5	0.0098	0.1948	4562.5	114.3	0.0065	0.1674	7557.8	130.4
Fsparse	0.0349	0.0889	3320.1	16.9	0.0372	0.0919	4562.9	20.1	0.0589	0.1116	7558.1	24.2
TV-MVP	**0.0012**	**0.0227**	1874.5	**1.2**	**0.0024**	**0.0225**	**1875.3**	**1.3**	**0.0050**	**0.0222**	**2587.8**	**1.6**

[a] The reported statistics are estimation errors of the risk, weight, covariance, and covariance of residuals.

[b] The number of factors in POET, Fglasso, Fsparse, and TV-MVP are all set to 2.

[c] The results are the average level of time points T, $T+25$ and $T+50$ for $p=50$, and T, $T+50$ and $T+100$ for $p=100$, $T+100$ and $T+200$ for $p=200$.

[d] The bandwidth of TV-MVP uses the rule of thumb for estimating time-varying factor loadings, and λ is selected by BIC.

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Table 2: Simulation results based on DGP 2

	p = 50, T = 100				**T = 200**				**T = 300**			
	Risk	Weight	Cov	Σ_e	Risk	Weight	Cov	Σ_e	Risk	Weight	Cov	Σ_e
Sample	0.0268	0.5159	491.5	-	0.0461	0.3762	569.4	-	0.0674	0.3403	624.0	-
Linear	0.0809	0.3272	503.4	-	0.0859	0.3260	575.8	-	0.0930	0.3161	628.4	-
Nonlinear	0.0787	0.2537	494.1	-	0.0900	0.2735	570.7	-	0.0941	0.2776	624.9	-
Nodewise	0.0662	0.2528	856.3	-	0.0704	0.2624	916.0	-	0.0681	0.2608	967.0	-
DCC-NL	0.0346	0.3993	436.3	-	0.0415	0.4060	421.0	-	0.0423	0.3903	270.3	-
POET	0.0179	0.1549	494.2	2.3	0.0180	0.1312	570.1	2.6	0.0216	0.1308	624.2	2.6
Fglasso	0.0287	0.4500	487.5	10.6	0.0494	0.3623	565.7	11.6	0.0688	0.3337	622.3	11.5
Fsparse	**0.0178**	0.1550	487.1	2.3	0.0181	0.1312	565.0	2.7	0.0216	0.1307	621.5	2.6
TV-MVP	0.0200	**0.1080**	**363.3**	**1.3**	**0.0121**	**0.0944**	**309.5**	**0.9**	**0.0159**	**0.0924**	**241.2**	**0.7**
	p = 100, T = 200				**T = 300**				**T = 400**			
	Risk	Weight	Cov	Σ_e	Risk	Weight	Cov	Σ_e	Risk	Weight	Cov	Σ_e
Sample	0.0146	0.3839	992.0	-	0.0246	0.3155	1005.6	-	0.0353	0.2930	1097.3	-
Linear	0.0331	0.2908	1003.1	-	0.0402	0.2807	1013.6	-	0.0465	0.2745	1103.4	-
Nonlinear	0.0517	0.2107	994.4	-	0.0549	0.2201	1007.3	-	0.0583	0.2277	1098.6	-
Nodewise	0.0230	0.1609	1632.7	-	0.0249	0.1672	1684.3	-	0.0228	0.1623	1753.0	-
DCC-NL	0.0178	0.2745	573.7	-	0.0214	0.2858	630.8	-	0.0193	0.2904	526.2	-
POET	0.0077	0.1009	995.0	3.6	0.0076	0.1000	1007.5	4.5	0.0109	0.1044	1098.4	5.1
Fglasso	0.0179	0.3185	982.8	21.1	0.0301	0.2877	1002.1	25.2	0.0393	0.2757	1091.2	27.5
Fsparse	0.0076	0.1009	983.0	3.7	0.0077	0.1001	1002.2	4.6	0.0111	0.1047	1091.0	5.2
TV-MVP	**0.0063**	**0.0627**	**442.9**	**1.2**	**0.0023**	**0.0518**	**560.0**	**0.9**	**0.0039**	**0.0539**	**500.4**	**0.8**
	p = 200, T = 500				**T = 600**				**T = 800**			
	Risk	Weight	Cov	Σ_e	Risk	Weight	Cov	Σ_e	Risk	Weight	Cov	Σ_e
Sample	0.0079	0.2431	2151.3	-	0.0118	0.2242	2198.8	-	0.0160	0.2076	2452.0	-
Linear	0.0137	0.2187	2161.8	-	0.0165	0.2097	2207.6	-	0.0192	0.2007	2458.4	-
Nonlinear	0.0271	0.1587	2153.6	-	0.0272	0.1600	2200.7	-	0.0285	0.1635	2453.4	-
Nodewise	0.0137	0.1207	3603.9	-	0.0144	0.1232	3662.5	-	0.0135	0.1200	3825.4	-
DCC-NL	0.0157	0.2164	1128.5	-	0.0162	0.2283	925.1	-	0.0151	0.2269	**634.5**	-
POET	0.0036	0.0647	2154.4	5.4	0.0038	0.0626	2201.3	5.9	0.0048	0.0677	2453.5	6.4
Fglasso	0.0127	0.2066	2146.6	42.0	0.0158	0.1987	2195.6	45.7	0.0192	0.1915	2451.8	48.2
Fsparse	0.0036	0.0648	2147.3	5.5	0.0038	0.0626	2196.1	6.0	0.0048	0.0677	2451.9	6.4
TV-MVP	**0.0016**	**0.0395**	**1038.4**	**1.0**	**0.0009**	**0.0243**	**874.8**	**0.9**	**0.0018**	**0.0326**	681.0	**0.8**

See notes in Table 1.

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Table 3: Simulation results based on the real monthly data of S&P 500 component stocks (DGP 3)

	p = 50, T = 100					**T = 200**					**T = 300**				
	Risk	Weight	SR	Cov	Σ_e	Risk	Weight	SR	Cov	Σ_e	Risk	Weight	SR	Cov	Σ_e
Sample	3.08E-04	0.5673	0.4073	0.0561	-	2.85E-04	0.4479	0.3639	0.0652	-	4.44E-04	0.3922	0.4186	0.0772	-
Linear	1.87E-04	0.3810	0.4053	0.0528	-	3.20E-04	0.3975	0.3770	**0.0643**	-	4.60E-04	0.3617	0.4295	0.0771	-
Nonlinear	**1.22E-04**	0.2712	0.3960	**0.0523**	-	3.70E-04	0.3385	0.3661	0.0647	-	4.86E-04	0.3141	0.4255	0.0769	-
Nodewise	6.41E-04	0.3038	2.9149	0.1193	-	2.88E-04	0.2484	2.3609	0.1121	-	1.71E-04	0.2168	2.2048	0.1297	-
DCC-NL	1.59E-04	0.3376	0.3150	0.0860	-	2.04E-04	0.3480	0.4888	0.0654	-	2.23E-04	0.2915	0.6179	0.0722	-
POET	1.83E-04	**0.2711**	0.3618	0.0559	**6.26E-03**	2.37E-04	0.3489	0.3267	0.0647	**5.49E-03**	2.84E-04	0.3098	0.3767	0.0786	5.70E-03
Fglasso	2.21E-04	0.5186	**0.0725**	0.1027	1.96E-02	3.87E-04	0.4674	0.1454	0.0953	1.65E-02	5.59E-04	0.4220	0.2152	0.0924	1.61E-02
Fsparse	1.23E-04	0.3333	0.0878	0.1028	1.16E-02	4.21E-04	0.3829	0.1487	0.0949	1.12E-02	4.81E-04	0.3397	0.2059	0.0931	1.01E-02
TV-MVP	3.81E-04	0.3119	0.1000	0.1138	1.02E-02	**1.51E-04**	**0.1730**	**0.0878**	0.0650	5.96E-03	**8.08E-05**	**0.1194**	**0.1026**	**0.0721**	**5.66E-03**
	p = 100, T = 200					**T = 300**					**T = 400**				
	Risk	Weight	SR	Cov	Σ_e	Risk	Weight	SR	Cov	Σ_e	Risk	Weight	SR	Cov	Σ_e
Sample	**8.12E-05**	0.5916	0.5956	0.1244	-	3.21E-04	0.4946	0.5631	0.1499	-	2.11E-04	0.4596	0.3715	0.1218	-
Linear	1.32E-04	0.4666	0.5813	**0.1215**	-	3.60E-04	0.4285	0.5663	0.1502	-	2.30E-04	0.4236	0.3788	0.1175	-
Nonlinear	2.69E-04	0.3436	0.5482	0.1223	-	4.56E-04	0.3258	0.5523	0.1493	-	2.94E-04	0.3465	0.3749	0.1188	-
Nodewise	2.68E-04	0.2565	4.1863	0.2109	-	1.41E-04	0.1942	3.8610	0.2585	-	3.48E-04	0.2534	3.4411	0.1352	-
DCC-NL	1.16E-04	0.3646	0.7067	0.1415	-	1.88E-04	0.3348	0.8052	**0.1310**	-	**9.70E-05**	0.3793	0.5339	0.1418	-
POET	1.90E-04	0.3585	0.5396	0.1225	**0.0073**	3.49E-04	0.3302	0.5327	0.1507	0.0075	1.36E-04	0.3579	0.3495	0.1257	0.0078
Fglasso	1.87E-04	0.5600	0.1341	0.1936	0.0282	4.69E-04	0.5104	0.2097	0.1789	0.0276	3.46E-04	0.4837	0.1120	0.2099	0.0272
Fsparse	3.02E-04	0.4406	0.1683	0.1928	0.0181	5.17E-04	0.3804	0.2352	0.1810	0.0099	2.77E-04	0.3863	0.1242	0.2136	0.0080
TV-MVP	1.42E-04	**0.2082**	**0.1026**	0.1341	0.0099	**6.18E-05**	**0.1221**	**0.0812**	0.1440	**0.0068**	1.59E-04	**0.2334**	**0.0806**	**0.1008**	**0.0062**
	p = 200, T = 300					**T = 350**					**T = 400**				
	Risk	Weight	SR	Cov	Σ_e	Risk	Weight	SR	Cov	Σ_e	Risk	Weight	SR	Cov	Σ_e
Sample	**3.59E-05**	0.5469	0.4944	0.3493	-	7.99E-05	0.4571	0.4555	0.3437	-	4.48E-05	0.4405	0.3980	0.2871	-
Linear	7.46E-05	0.3917	0.4725	0.3476	-	9.45E-05	0.3696	0.4489	0.3445	-	3.86E-05	0.3806	0.3886	0.2758	-
Nonlinear	2.10E-04	0.2286	0.4472	0.3460	-	1.73E-04	0.2312	0.4365	0.3412	-	1.07E-04	0.2603	0.3659	0.2787	-
Nodewise	9.10E-05	0.1350	5.9352	0.5794	-	1.22E-04	0.1471	5.8878	0.6244	-	2.05E-04	0.1917	5.8505	0.2908	-
DCC-NL	8.70E-05	0.2498	0.7579	0.3609	-	8.04E-05	0.2515	0.8134	0.3516	-	4.68E-05	0.2852	0.6183	0.3483	-
POET	1.26E-04	0.2374	0.4754	0.3499	**0.0126**	1.12E-04	0.2131	0.5134	0.3446	0.0159	**2.66E-05**	0.2502	0.4267	0.2906	0.0183
Fglasso	9.68E-05	0.3897	0.0834	0.4356	0.0621	1.07E-04	0.3711	0.1000	0.4027	0.0631	5.37E-05	0.3796	0.0595	0.5271	0.0632
Fsparse	1.84E-04	0.2480	0.1583	0.4392	0.0143	1.39E-04	0.2242	0.1972	0.4062	0.0181	4.67E-05	0.2553	0.1318	0.5318	0.0184
TV-MVP	4.18E-05	**0.0913**	**0.0450**	**0.3152**	0.0146	**7.82E-05**	**0.0990**	**0.0412**	**0.3212**	**0.0126**	1.25E-04	**0.1646**	**0.0406**	**0.2532**	**0.0106**

[a] The reported statistics are estimation errors of the risk, weight, Sharpe ratio, covariance, and covariance of residuals.
[b] The number of factors in POET, Fglasso, Fsparse, and TV-MVP are all set to 2.
[c] The results are the average level of time points T, $T + 25$ and $T + 50$.
[d] The bandwidth of TV-MVP selects the rule of thumb, and λ is selected by BIC.

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Table 4: Simulation results based on the real daily data of S&P 500 component stocks (DGP 4)

	p = 50, T = 100					**T = 200**					**T = 300**				
	Risk	Weight	SR	Cov	Σ_e	Risk	Weight	SR	Cov	Σ_e	Risk	Weight	SR	Cov	Σ_e
Sample	6.10E-06	0.3092	0.1898	1.76E-03	-	4.13E-06	0.1803	0.0966	1.46E-03	-	1.62E-05	0.2200	0.0844	2.08E-03	-
Linear	1.79E-06	0.1521	0.1534	1.66E-03	-	2.67E-06	0.1344	0.0956	1.26E-03	-	1.53E-05	0.2117	0.0862	2.12E-03	-
Nonlinear	**1.48E-06**	0.1269	0.1449	**1.53E-03**	-	**1.88E-06**	0.1199	0.0927	1.27E-03	-	**1.39E-05**	0.2144	0.0844	2.05E-03	-
Nodewise	9.79E-06	0.1566	0.5211	3.19E-03	-	1.09E-05	0.1513	0.3892	2.29E-03	-	2.72E-05	0.2682	0.3818	3.78E-03	-
DCC-NL	1.92E-06	0.1314	0.1407	1.55E-03	-	2.72E-06	0.1139	0.0897	**1.12E-03**	-	1.48E-05	0.2011	0.0767	1.93E-03	-
POET	3.53E-06	0.1085	0.1476	1.55E-03	3.93E-04	4.78E-06	0.1043	0.0977	1.52E-03	3.30E-04	1.88E-05	0.2180	0.0887	2.09E-03	3.07E-04
Fglasso	5.36E-06	0.2571	0.0833	1.78E-03	1.18E-03	3.96E-06	0.1679	0.0664	1.52E-03	8.85E-04	1.61E-05	0.2145	0.0640	2.04E-03	7.65E-04
Fsparse	2.63E-06	**0.1065**	0.1042	1.60E-03	3.98E-04	4.03E-06	0.0982	0.0785	1.60E-03	**3.15E-04**	1.80E-05	0.2124	0.0748	2.07E-03	2.92E-04
TV-MVP	3.07E-06	0.1082	**0.0547**	2.31E-03	**3.92E-04**	4.88E-06	**0.0970**	**0.0419**	1.84E-03	3.20E-04	1.64E-05	**0.1969**	**0.0476**	**1.87E-03**	**2.91E-04**
	p = 100, T = 200					**T = 300**					**T = 400**				
	Risk	Weight	SR	Cov	Σ_e	Risk	Weight	SR	Cov	Σ_e	Risk	Weight	SR	Cov	Σ_e
Sample	6.00E-06	0.2557	0.1403	2.86E-03	-	1.07E-05	0.2767	0.1047	4.34E-03	-	7.99E-06	0.2431	0.0894	7.98E-03	-
Linear	3.74E-06	0.1544	0.1161	2.36E-03	-	9.23E-06	0.2442	0.1029	4.46E-03	-	6.80E-06	0.2248	0.0907	8.29E-03	-
Nonlinear	**2.49E-06**	0.1234	0.1056	2.40E-03	-	**7.78E-06**	0.2408	0.0971	4.27E-03	-	**4.60E-06**	0.2199	0.0886	8.07E-03	-
Nodewise	8.98E-06	0.1503	0.5550	3.85E-03	-	1.82E-05	0.2730	0.5380	7.42E-03	-	2.03E-05	0.3089	0.5785	1.34E-02	-
DCC-NL	3.13E-06	0.1173	0.0970	**1.94E-03**	-	8.15E-06	0.2220	0.0851	3.87E-03	-	6.19E-06	0.2042	0.0798	6.78E-03	-
POET	4.52E-06	0.1084	0.1104	2.66E-03	4.55E-04	1.14E-05	0.2484	0.1039	4.24E-03	4.19E-04	9.28E-06	0.2204	0.0938	7.81E-03	4.46E-04
Fglasso	5.34E-06	0.2028	0.0623	2.95E-03	1.76E-03	1.02E-05	0.2559	0.0640	4.27E-03	1.49E-03	7.52E-06	0.2259	0.0628	7.80E-03	1.44E-03
Fsparse	4.21E-06	0.1061	0.0836	2.80E-03	**4.37E-04**	1.10E-05	0.2453	0.0842	4.19E-03	**4.02E-04**	8.82E-06	0.2170	0.0772	7.66E-03	4.27E-04
TV-MVP	4.92E-06	**0.1035**	**0.0388**	3.12E-03	4.45E-04	9.56E-06	**0.2185**	**0.0487**	**3.86E-03**	4.07E-04	6.64E-06	**0.1611**	**0.0459**	**5.74E-03**	**4.17E-04**
	p = 200, T = 500					**T = 600**					**T = 800**				
	Risk	Weight	SR	Cov	Σ_e	Risk	Weight	SR	Cov	Σ_e	Risk	Weight	SR	Cov	Σ_e
Sample	1.21E-06	0.2471	0.0855	0.0901	-	1.87E-06	0.2370	0.0666	0.1273	-	6.84E-06	0.2483	0.0517	0.0789	-
Linear	7.82E-07	0.2058	0.0808	0.0909	-	2.50E-06	0.2109	0.0636	0.1282	-	7.56E-06	0.2340	0.0503	0.0787	-
Nonlinear	3.82E-06	0.1224	0.0743	0.0904	-	5.81E-06	0.1331	0.0559	0.1276	-	1.03E-05	0.1706	0.0471	0.0789	-
Nodewise	9.01E-06	0.1673	1.0319	0.1182	-	8.73E-06	0.1634	1.1596	0.1742	-	6.53E-06	0.1393	1.3303	0.0989	-
DCC-NL	1.89E-06	0.1576	0.1237	**0.0531**	-	2.88E-06	0.1984	0.1516	**0.0678**	-	2.77E-06	0.2399	0.2132	**0.0410**	-
POET	1.19E-06	0.1034	0.0811	0.0896	8.76E-04	1.25E-06	0.1070	0.0774	0.1268	8.77E-04	2.48E-06	0.1232	0.0839	0.0790	9.04E-04
Fglasso	7.91E-07	0.2000	0.0473	0.0895	2.96E-03	2.49E-06	0.2015	0.0414	0.1265	2.92E-03	7.64E-06	0.2239	0.0367	0.0794	3.07E-03
Fsparse	**6.95E-07**	0.0969	0.0654	0.0890	8.01E-04	1.20E-06	0.0970	0.0656	0.1261	7.98E-04	3.12E-06	0.1139	0.0736	0.0795	7.72E-04
TV-MVP	1.07E-06	**0.0945**	**0.0261**	0.0744	**7.90E-04**	**1.14E-06**	**0.0918**	**0.0182**	0.1004	**7.59E-04**	**2.45E-06**	**0.1110**	**0.0097**	0.0767	**7.22E-04**

See notes in Table 1. The reported statistics include the Sharpe ratio, besides the four measures in Table 1.

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factor loadings, it returns to the average level at some time points (e.g., after the time point 400). At these points, the time-varying factor loading happens to take value close to the constant factor loading model. Also, from the right subfigure of Figure 1, it is evident that during the period 1-400, there are few variations in factor loadings. However, from another perspective, the results of the TV-MVP strategy are robust to the "timing" of variations in population factor loadings.

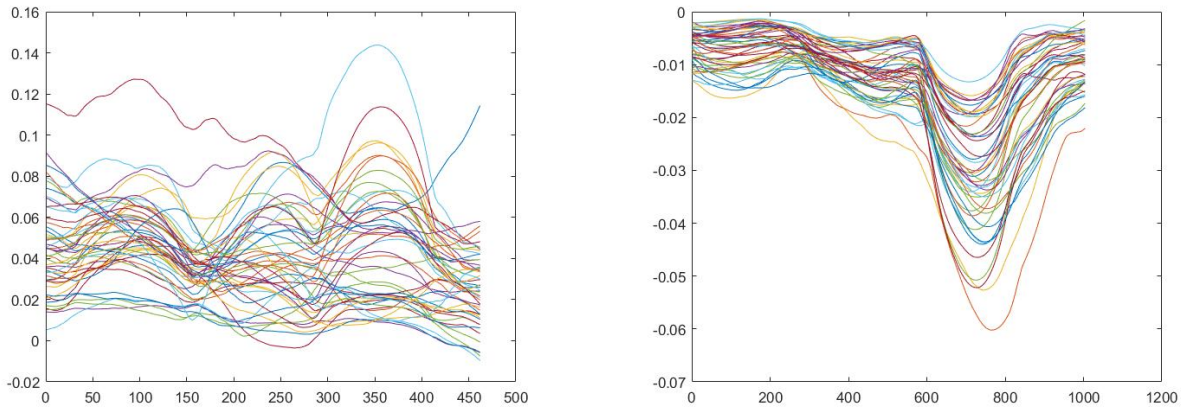

Figure 1: Population time-varying factor loadings in DGP 3 with $p = 50$

6.2 Simulation for Hypothesis Test

6.2.1 Model Set-up

In this part, we let $\mu_{f1} = 0$ and $\mu_{f2} = 0$, and consider the following eight DGPs in generating factor loadings and residuals.

1. Size1 (i.i.d. in residuals): $b_i \overset{\text{i.i.d.}}{\sim} N(0, I_2)$ and $e_i \overset{\text{i.i.d.}}{\sim} N(0, 1)$

2. Size2 (Heteroskedasticity in residuals): $b_i \overset{\text{i.i.d.}}{\sim} N(0, I_2)$, $e_i = \sigma_i v_i$ where $v_i \overset{\text{i.i.d.}}{\sim} N(0, 1)$ and $\sigma_i \overset{\text{i.i.d.}}{\sim} \text{Unif}(0.5, 1.5)$.

3. Size3 (Cross-sectional dependence in residuals): $b_i \overset{\text{i.i.d.}}{\sim} N(0, I_2)$, $e_t \sim N(0, \Sigma_e)$ for all t, where $\Sigma_e = (a_{ij})_{i,j=1,\ldots p}$ and $a_{ij} = 0.5^{|i-j|}$.

4. Power1-step2 (Change point in residual covariance): $b_{it} \overset{\text{i.i.d.}}{\sim} N(0, I_2)$, $e_{it} \overset{\text{i.i.d.}}{\sim} N(0, 1)$ for $1 \leq t \leq T/2$, and $e_{it} \overset{\text{i.i.d.}}{\sim} N(0, 1 + \rho)$ for $T/2 + 1 \leq t \leq T$.

5. Power2-step2 (Smooth changes in residual covariance): $b_{it} \overset{\text{i.i.d.}}{\sim} N(0, I_2)$, and $e_{it} \overset{\text{i.i.d.}}{\sim} N(0, (1 + \frac{3\rho t}{T})\Sigma_e)$ where $\Sigma_e = (a_{ij})_{i,j=1,\ldots d}$ and $a_{ij} = 0.5^{|i-j|}$.

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6. Power3-step1 (Step function in factor loadings): b_{it} is the values of step function with step length D:

$$b_{it}^{(1)} = \begin{cases} 2|z_i^{(1)}| & \text{for} \quad 1 \le t \le D \\ 2|z_i^{(1)}| + \rho z_i^{(2)} & \text{for} \quad D+1 \le t \le 2D \\ 2|z_i^{(1)}| + \rho z_i^{(2)} + \rho z_i^{(3)} & \text{for} \quad 2D+1 \le t \le 3D \\ \vdots & \\ 2|z_i^{(1)}| + \rho \sum_{j=2}^{k} z_i^{(j)} & \text{for} \quad (k-1)D+1 \le t \le kD \\ \vdots & \end{cases}$$

where $z_i^{(j)}$ for $j = 1, 2, \ldots, 6$, are all standard normal random variables. Here we fix the number of steps with six and $D = [T/6]$ where operator $[\cdot]$ takes the integer part of a real number. $b_{it}^{(2)}$ has the same form with replacing $z_i^{(j)}$ by $|v_i^{(j)}|$ and $v_i^{(j)} \sim N(0,1)$ for $j = 1, 2, \ldots, 6$. This DGP is an adaptation of the DGP 1 in subsection 6.1.1 by adding the parameter ρ, which is used for calculating local power.

7. Power4-step1 (Smooth changes in factor loadings): Similar to Power3-step1, this DGP is the adaptation of the DGP 2 in subsection 6.1.1 by adding ρ, that is

$$b_{it}^{(1)} = \rho \left((3 + z_i^{(1)})t/T + z_i^{(2)} sin(4\pi t/T) \right) + z_i^{(3)}$$

where $z^{(j)}, j = 1, 2, 3$ are standard normal variables. And similarly,

$$B_{it}^{(2)} = \rho \left((3 + z_i^{(1)})t/T + z_i^{(2)} sin(4\pi t/T) + (z_i^{(4)} t/T)^2 \right) + z_i^{(3)}$$

and $e_{it} \overset{\text{i.i.d.}}{\sim} N(0,1)$.

8. Power5 (Step changes in factor loadings and change point in residual covariance): b_{it} and e_{it} in this DGP are generated in the same way with Power3-step1 and Power1-step2, respectively, to explore the power of situations where both factor loadings and residual covariance are time-varying.

DGPs 1-3 satisfy the null hypothesis of time-invariant factor loadings and residual covariance matrices. These three DGPs are used to study the size in finite sample. DGP 1 is the most simple case with residuals from i.i.d. standard multivariate normal distribution, DGP 2 further allows the time-invariant heteroskedasticity for individual units of residuals. DGP 3 allows the time-invariant cross-sectional dependence among residuals. DGPs 4 and 5 are designed to show the power of the second step test. Specifically, DGP 4 exhibits the change point in the residual covariance matrix; DGP5 considers a smooth linear increase in cross-section dependent type residual covariance matrix. A parameter ρ controlling the level of instability is added. DGP 6 and DGP 7 are designed to show the power of the step 1 test with sudden and smooth changes in factor loadings, respectively. At last, DGP8 has time-varying factor loadings and a time-varying residual covariance matrix.

For the first step of our sequential test, we conduct the bootstrap version of the test, which is less sensitive to the selection of bandwidth, more precisely,

1. With the data Y_t, we first apply the conventional PCA and local PCA to get the time-invariant and variant factor loadings $\widetilde{b}_i, \widehat{b}_{it}$ and corresponding common factors $\widetilde{F}_t, \widehat{F}_t$. Then the statistic $\widehat{J}_{pT}$ can

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be calculated by (5.5). The Epanechnikov kernel and the rule of thumb bandwidth $h = \frac{2.35}{\sqrt{12}} T^{-1/5} p^{-1/10}$ are applied during the test.

2. Generate the bootstrap residuals by $e_t^* = \widetilde{\Sigma}_e^{1/2} v_t$ where $v_t = (v_{1t}, ..., v_{pt})^\top$, v_{it} is from i.i.d. $N(0,1)$ and $(\widetilde{\Sigma}_e)_{ij} = (S_e)_{ij}(1-\iota)^{|i-j|}$, ι is a small positive number (e.g. 0.01 in our simulation) and S_e is sample covariance matrix of residuals from conventional PCA.

3. Generate the bootstrap data $Y_t^* = \widetilde{B}\widetilde{F} + e_t^*$, and follow the same procedure in step 1 to obtain $\widehat{J}_{pT}^*$ using Y_t^*.

4. Repeat steps 2 and 3 for N_b times and the bootstrap p-value is given by $p^* = \frac{1}{N_b}\sum_{i=1}^{N_b} \mathbb{1}\{\widehat{J}_{pT}^* \geq \widehat{J}_{pT}\}$.

Table 5: Size of the tests under DGP 1-3

DGP	p	T	5%			10%		
			Step 1	Step 2*	Total	Step 1	Step 2	Total
size1	30	200	0.022	15/489	0.052	0.052	19/474	0.090
	50	300	0.012	23/494	0.058	0.038	32/481	0.102
	100	300	0.020	20/490	0.060	0.038	25/481	0.088
	150	300	0.026	14/487	0.054	0.046	21/456	0.088
	200	400	0.028	21/486	0.070	0.044	23/478	0.090
size2	30	200	0.028	16/486	0.060	0.050	24/475	0.098
	50	300	0.018	17/491	0.052	0.042	25/479	0.092
	100	300	0.028	21/486	0.070	0.050	31/475	0.112
	150	300	0.028	9/486	0.046	0.046	14/477	0.074
	200	400	0.020	18/490	0.056	0.048	25/476	0.098
size3	30	200	0.014	13/493	0.040	0.046	18/477	0.082
	50	300	0.014	16/493	0.046	0.046	21/477	0.088
	100	300	0.018	10/491	0.038	0.048	17/476	0.082
	150	300	0.020	22/490	0.064	0.038	27/481	0.092
	200	400	0.014	14/493	0.042	0.048	19/476	0.086

$*$ If we reject the constancy in the first step, it is a rejection of the overall null. If we do not reject the first step but reject it in the second step, it also rejects the overall null. To show how many times (in the 500 simulations) the nulls are *not rejected* (that is, the denominator) in the first step and are consequently rejected (the numerator) in the second step, we use a fraction to report the size of step 2.

6.2.2 Results

Table 5 shows that, in different combinations of portfolio and sample sizes, the overall test is close to the nominal level. Furthermore, when the sample size T gets larger, the empirical sizes of the overall test are getting closer to the nominal sizes. In general, in empirically relevant cases, our test can control the empirical size well.

Our test is powerful in detecting all the forms of time-varying factor loadings and residual covariance matrices as shown in Table 6. The test of Su and Wang (2017) can only detect the time-varying in factor loadings from DGP 6 to DGP 8. One advantage of our test is that it can clarify the time-varying parts in the decomposed covariance matrix, which could help the users to choose the appropriate method. Furthermore, we explore the ability of our test to detect the departures from the null hypothesis.

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Table 6: Power of tests of under DGP4-8

DGP	d	T	5%			10%		
			Step1	Step2	Total	Step1	Step2	Total
Power1	30	50	0.004	217/498	0.438	0.016	266/492	0.548
$\rho = 0.25$	50	100	0.018	486/491	0.990	0.032	483/484	0.998
	100	150	0.014	493/493	1.000	0.032	484/484	1.000
Power1	30	50	0.010	484/495	0.978	0.014	489/493	0.992
$\rho = 0.5$	50	100	0.012	494/494	1.000	0.030	485/485	1.000
	100	150	0.018	491/491	1.000	0.028	486/486	1.000
Power1	30	50	0.006	497/497	1.000	0.010	495/495	1.000
$\rho = 1$	50	100	0.008	496/496	1.000	0.014	493/493	1.000
	100	150	0.016	492/492	1.000	0.046	477/477	1.000
Power2	30	50	0.006	274/497	0.554	0.018	307/491	0.632
$\rho = 0.25$	50	100	0.006	495/497	0.996	0.018	489/491	0.996
	100	150	0.018	491/491	1.000	0.032	484/484	1.000
Power2	30	50	0.004	454/498	0.912	0.022	451/489	0.924
$\rho = 0.5$	50	100	0.014	493/493	1.000	0.040	480/480	1.000
	100	150	0.022	489/489	1.000	0.044	478/478	1.000
Power2	30	50	0.010	474/495	0.958	0.016	483/492	0.982
$\rho = 1$	50	100	0.028	486/486	1.000	0.054	473/473	1.000
	100	150	0.020	490/490	1.000	0.040	480/480	1.000
Power3	30	50	0.958	0	0.958	0.972	0	0.972
$\rho = 0.25$	50	100	1.000	0	1.000	1.000	0	1.000
	100	150	1.000	0	1.000	1.000	0	1.000
Power3	30	50	1.000	0	1.000	1.000	0	1.000
$\rho = 0.5$	50	100	1.000	0	1.000	1.000	0	1.000
	100	150	1.000	0	1.000	1.000	0	1.000
Power3	30	50	1.000	0	1.000	1.000	0	1.000
$\rho = 1$	50	100	1.000	0	1.000	1.000	0	1.000
	100	150	1.000	0	1.000	1.000	0	1.000
Power4	30	50	0.976	0	0.976	0.988	0	0.988
$\rho = 0.25$	50	100	1.000	0	1.000	1.000	0	1.000
	100	150	1.000	0	1.000	1.000	0	1.000
Power4	30	50	0.996	1/2	0.998	1.000	0	1.000
$\rho = 0.5$	50	100	1.000	0	1.000	1.000	0	1.000
	100	150	1.000	0	1.000	1.000	0	1.000
Power4	30	50	0.982	4/9	0.990	1.000	0	1.000
$\rho = 1$	50	100	1.000	0	1.000	1.000	0	1.000
	100	150	1.000	0	1.000	1.000	0	1.000
Power5	30	50	0.910	10/45	0.930	0.948	10/26	0.968
$\rho = 0.25$	50	100	1.000	0	1.000	1.000	0	1.000
	100	150	1.000	0	1.000	1.000	0	1.000
Power5	30	50	0.996	2/2	1.000	0.996	2/2	1.000
$\rho = 0.5$	50	100	1.000	0	1.000	1.000	0	1.000
	100	150	1.000	0	1.000	1.000	0	1.000
Power5	30	50	1.000	0	1.000	1.000	0	1.000
$\rho = 1$	50	100	1.000	0	1.000	1.000	0	1.000
	100	150	1.000	0	1.000	1.000	0	1.000

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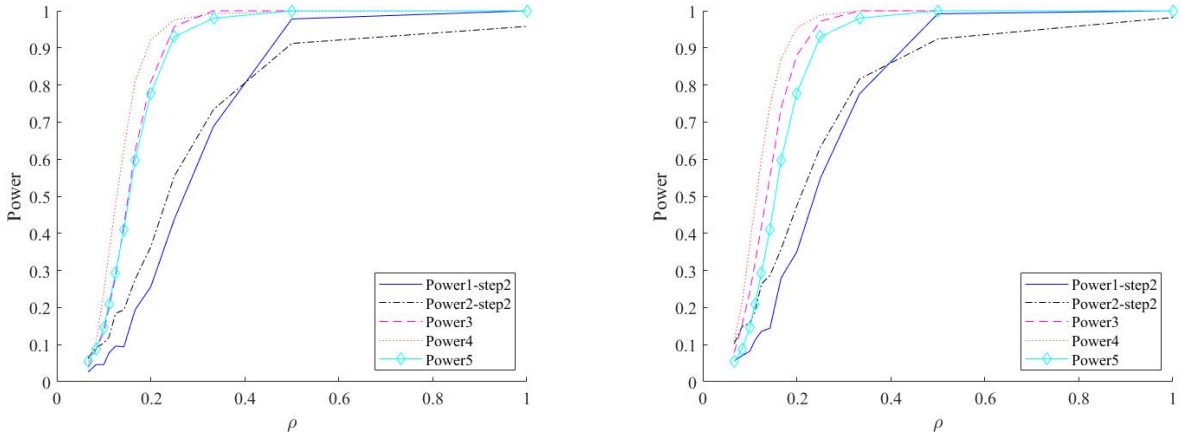

Figure 2: Local power for DGP4-8 with $p = 30$ and $T = 50$ under the confident level 5%(left) and 10%(right) with respect to varying ρ.

Figure 2 shows the local power of our test with respect to varying ρ with $p = 30$ and $T = 50$. The figure shows that the power increases with ρ for all alternatives, reaching 1 with reasonably large ρ.

7 An empirical application to MVP using S&P 500 stocks

In this section, we demonstrate that the proposed TV-MVP has better out-of-sample performance in terms of risk and Sharpe ratio than other investment strategies. We focus on three portfolio performance measures: Sharpe ratio, risk (standard deviation), and cumulative excess return. The competing strategies include those mentioned in the simulations section, plus the benchmark "1/N" and the unified minimum variance problem estimator (MVP-UF) proposed by Ding et al. (2021), which is risk scenario-adaptive.

At the beginning of the investment decision period (say T_0), we use the historical (before T_0) excess return data (the training data set) with length T to compute covariance matrix estimators and optimal weights of a portfolio. We consider $T = 125, 250$ in this section. The holding period for all portfolios is HT. To mimic real-world mutual fund practice, we use the holding time for one week (5 trading days) and one month (21 trading days). During the holding period, we use excess return data r_t to calculate the returns of each portfolio. After the holding period, we re-construct the portfolios using historical data of the same length T in a rolling window manner. We repeat this process until the last period of the dataset. We assume the same holding period for all strategies and no transaction fees or fund fees to make a fair comparison.

Specifically, the aforementioned performance measures are:

$$\text{Cumulative excess return} = \sum_{l=1}^{RT} \sum_{t=1}^{HT} \widehat{W}_l^{\top} r_t^l \tag{7.1}$$

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$$\text{Standard deviation} = \sqrt{\frac{1}{RT \times HT - 1} \sum_{l=1}^{RT} \sum_{t=1}^{HT} \left(\widehat{W_l}^{\top} r_t^l - \frac{\text{Cumulative excess return}}{RT \times HT} \right)^2} \tag{7.2}$$

$$\text{SR} = \frac{1}{RT \times HT} \frac{\text{Cumulative excess return}}{\text{Standard deviation}} \tag{7.3}$$

where RT is the number of rebalancing events, r_t^l is the p-dimensional realized excess return in l-th rebalancing, p is the number of assets in investment universe, and $\widehat{W_l}$ is the optimal weight estimator in l-th rebalancing process. When the last holding period is less than HT, we assume the fund manager will hold the assets for any time until the end of the investment period.

In this section, we use the method of Su and Wang (2017) to determine the number of common factors $\hat{m}$. For the bandwidth choice in practice, one could use the rule of thumb as in the main text. We recommend using a data-driven method to determine the bandwidth. Specifically, we apply a simple time series split cross-validation technique (Varma and Simon, 2006) with the target of Sharpe ratio. The details of this cross-validation method are put in the appendix. Furthermore, for the values of λ in the residual covariance matrix estimation, we also use cross-validation to determine its value.

7.1 Data

We use daily return data of S&P 500 index component stocks downloaded from CRSP. Here different holding times and sample stocks (randomly selecting stocks from S&P 500 or the largest market cap) are considered to evaluate the robustness of portfolio performance. We drop the corresponding assets from the basket following Ao et al. (2019) to avoid look-ahead bias for the missing data during the aforementioned rolling window process. Finally, for risk-free return data, we take it from the Fama-Frech data library.

We conduct the empirical study based on the following scenarios: (1) The out-of-sample period is from 03/01/1995 to 12/31/2019. Portfolio size $p = 50, 100$, sample size $T = 125$ (half-year). (2) The out-of-sample period is from 03/01/2006 to 31/12/2015. Portfolio size $p = 100$, sample size $T = 125, 250$ (half-year, one year). The holding period HT is set to 21 days (monthly position rebalance) for the first two scenarios. (3) The out-of-sample period is from 03/01/2006 to 31/12/2015. Portfolio size $p = 100$, sample size $T = 125, 250$. (4) The out-of-sample period is from 04/01/2010 to 12/31/2019. Portfolio size $p = 50$, sample size $T = 125, 250$. The holding period HT is set to 5 days (weekly position rebalance) for the last two scenarios.

Notice that we update factor number estimator $\hat{m}$, λ, and bandwidth h annually for TV-MVP. Moreover, the common factor number is set to 3 for other factor-based strategies, which are also suggested using the method of Bai and Ng (2002).

We first performed the hypothesis testing of the constant covariance matrix in the studied sample period on a rolling window basis. The null hypothesis was rejected around 60% of the time under 10% confidence level. And most of the time (70%), it was rejected in step 1, but not in step 2. This result is robust for different (p, T) combinations shown in Table 7.

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7.2 Results

The empirical results are shown in Table 7. It clearly shows our strategy has good performance in practice. In both the monthly and weekly rebalancing cases, the TV-MVP is among the portfolios with the best returns. More importantly, TV-MVP is the best overall performer in terms of Sharpe ratio. It achieves a good balance between satisfactory returns while the risk is controlled at the near-minimum level.

8 Conclusion

In this paper, we study a novel time-varying minimum variance portfolio, whose covariance is allowed to have structural changes over time. Under the time-varying factor structure on asset returns, a factor-augmented quasi-likelihood estimation approach is proposed to obtain a more efficient estimator for the time-varying covariance structure for asset returns. We also propose a test of the constancy of the covariance matrix. The TV-MVP is shown to have desired theoretical properties, including sharp risk consistency. Simulation and empirical studies provide further evidence that the investment strategy using TV-MVP has a very satisfactory performance.

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Table 7: Empirical daily application of holding stocks in S&P index

	Return	Risk	SR	Return	Risk	SR	Return	Risk	SR	Return	Risk	SR
	1995-2019, monthly rebalancing						2010-2019, weekly rebalancing					
	p = 50, T = 125			p = 100, T = 125			p = 50, T = 250			p = 50, T = 125		
1/N	3.059	1.40E-04	0.0412	3.179	1.41E-04	0.0428	1.268	1.10E-04	0.0481	1.445	1.12E-04	0.0544
Sample	2.999	9.63E-05	0.0487	2.736	1.02E-04	0.0433	1.527	5.11E-05	0.0849	1.710	6.17E-05	0.0866
Linear	2.707	7.95E-05	0.0484	2.256	6.26E-05	0.0456	1.500	4.75E-05	0.0865	1.619	5.02E-05	0.0909
Non-linear	2.582	7.94E-05	0.0461	2.114	5.95E-05	0.0438	1.544	4.69E-05	0.0897	1.429	4.61E-05	0.0837
nodewise	2.803	9.41E-05	0.0460	2.848	9.34E-05	0.0471	1.234	6.74E-05	0.0598	1.437	7.50E-05	0.0660
POET	1.889	8.04E-05	0.0336	2.230	6.50E-05	0.0442	1.486	6.00E-05	0.0763	1.370	5.03E-05	0.0768
MVP_UF	2.231	7.75E-05	0.0404	2.273	6.14E-05	0.0464	1.123	5.01E-05	0.0631	1.282	5.07E-05	0.0716
fglasso	3.011	9.53E-05	0.0491	2.666	9.60E-05	0.0435	1.501	5.08E-05	0.0837	1.648	6.12E-05	0.0838
DCC-NL	2.586	8.08E-05	0.0458	2.089	6.04E-05	0.0429	1.440	4.99E-05	0.0810	1.494	4.62E-05	0.0874
TV-MVP	3.229	8.61E-05	0.0554	2.390	5.88E-05	0.0498	1.734	5.56E-05	0.0924	1.749	4.97E-05	0.0987
	2006-2015, monthly rebalancing						2006-2015, weekly rebalancing					
	p = 100, T = 250			p = 100, T =125			p = 100, T = 125			p = 100, T= 250		
1/N	0.936	2.11E-04	0.0257	1.035	2.19E-04	0.0280	0.919	2.11E-04	0.0251	1.103	2.28E-04	0.0291
Sample	0.833	8.49E-05	0.0362	1.225	1.25E-04	0.0439	1.492	1.75E-04	0.0449	1.037	1.03E-04	0.0407
Linear	0.801	7.21E-05	0.0377	0.850	2.93E-04	0.0199	1.078	7.35E-05	0.0500	0.988	9.54E-05	0.0402
Non-linear	0.794	6.93E-05	0.0382	1.129	6.76E-05	0.0549	0.983	6.84E-05	0.0473	0.974	9.34E-05	0.0401
nodewise	0.905	1.32E-04	0.0315	1.058	1.44E-04	0.0353	0.912	1.35E-04	0.0312	1.059	1.45E-04	0.0350
POET	0.828	8.47E-05	0.0360	0.793	6.57E-05	0.0392	0.995	7.57E-05	0.0455	0.960	9.90E-05	0.0384
MVP_UF	0.625	8.18E-05	0.0276	0.849	7.09E-05	0.0403	0.819	7.38E-05	0.0379	0.760	1.03E-04	0.0297
fglasso	0.808	8.41E-05	0.0353	1.127	1.11E-04	0.0428	1.407	1.39E-04	0.0475	1.030	1.02E-04	0.0405
DCC-NL	1.084	7.16E-05	0.0513	1.154	6.40E-05	0.0577	0.946	6.87E-05	0.0454	0.766	8.54E-05	0.0330
TV-MVP	1.135	7.43E-05	0.0527	1.158	6.83E-05	0.0561	1.093	7.36E-05	0.0507	1.025	8.11E-05	0.0453

[a] We report the cumulative excess return, the standard deviation of portfolio return, and the Sharpe ratio defined in the main text.
[b] Left part in each sub-block is largest cap assets, and right part in each sub-block is randomly selected assets.
[c] Data-driven cross-validation methods are used for λ and h of TV-MVP, and information criterion in Su and Wang (2017) is used for $\hat{m}$.

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Supplementary Material to "Time-varying Minimum Variance Portfolio"

Qingliang Fan[†], Ruike Wu[‡], Yanrong Yang[♯], and Wei Zhong[‡]
Department of Economics, The Chinese University of Hong Kong[†]
WISE and School of Economics, Xiamen University[‡]
College of Business and Economics, The Australian National University[♯]

In this supplementary material, we provide three parts: Appendix A show the details of computation algorithms; Appendix B present all proofs for main theorems; some lemmas and related proofs are provided in Appendix C.

9 Appendix A

This appendix provides the details of computation algorithms, including the shrinkage method in quasi-likelihood estimation for Σ_e, the choice of bandwidth in kernel estimation for time-varying factor model, determining the number of factors in the time-varying factor model, and some modified assumptions without the sub-Gaussian condition.

9.1 Details of the shrinkage estimation of Σ_e in Section 3.2

First, a function $f(x)$ is said to be majorized by $g(x|\,x_0)$, if $f(x) \leq g(x|\,x_0)$ for all x and $f(x_0) = g(x_0\,|\,x_0)$. For the objective function (3.4), which has the form of the sum of a convex and a concave function, a commonly used majorizer replaces the concave function's part with its tangent. The concave part of the objective function (3.4) $\log\det(\Sigma_e) \leq \log\det(\Sigma_{e,0}) + \mathrm{tr}(\Sigma_{e,0}^{-1}(\Sigma_e - \Sigma_{e,0}))$ where $\Sigma_{e,0}$ is some known positive-definite matrix. Therefore, function (3.4) can be majorized by $g(\Sigma_e|\,\Sigma_{e,0}) = \log\det(\Sigma_{e,0}) + \mathrm{tr}(\Sigma_{e,0}^{-1}\,\Sigma_e) - p + \mathrm{tr}(\Sigma_e^{-1}\,S_e) + \|\Lambda * \Sigma_e\|_1$. Then, we can solve Equation (3.4) by iteratively solving Equation (9.1).

$$\widehat{\Sigma}_e^{(t)} = \underset{\Sigma_e \succ 0}{\arg\min}\, \mathrm{tr}\left(\left(\widehat{\Sigma}_e^{(t-1)}\right)^{-1} \Sigma_e\right) + \mathrm{tr}\left(\Sigma_e^{-1}\,S_e\right) + \|\Lambda * \Sigma_e\|_1 \tag{9.1}$$

Equation (9.1) is convex, a generalized gradient descent algorithm can easily solve it, and a global minimum is guaranteed. Note that, in the following computation, the positive-definite constraint $\Sigma_e \succ 0$ is tightened to $\Sigma_e \succeq \delta I$ for $\delta > 0$ if $S_e \succ 0$, which depends on the smallest eigenvalue of S_e. In detail, let $L(\Sigma_e) = \mathrm{tr}(\left(\widehat{\Sigma}_e^{(t-1)}\right)\Sigma_e) + \mathrm{tr}(\Sigma_e^{-1}S_e)$ which is the differentiable part of Equation (9.1). Then $dL(\Sigma_e)/d\Sigma_e = \widehat{\Sigma}_e^{(t-1)} - \Sigma_e^{-1}S_e\Sigma_e^{-1}$. Thus the update steps of the generalized gradient descent algorithm are given by

$$\Sigma_e = \arg\min_{\Xi_e \succeq \delta I} (2\gamma)^{-1} \left\|\Xi_e - \Sigma_e + \gamma\left(\widehat{\Sigma}_e^{(t-1)} - \Sigma_e^{-1}S_e\Sigma_e^{-1}\right)\right\|_F^2 + \|\Lambda * \Xi_e\|_1 \tag{9.2}$$

where γ is the step size in each iteration which is generally calculated by the backtracking line search approach. Note that the corresponding optimization problem without constraint $\Xi_e \succeq \delta I$ is easier and

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has the solution shown by Equation (9.3),

$$\Sigma_e \leftarrow ST\{\Sigma_e - \gamma((\widehat{\Sigma}_e^{(t-1)})^{-1} - \Sigma_e^{-1} S_e \, {\Sigma_e}^{-1}), \Lambda\gamma\} \tag{9.3}$$

where $ST(A,B)_{ij} = \text{sgn}(A_{ij})\,(\mid A_{ij} \mid - B_{ij})_+$, which is a soft-thresholding operator ($\text{sgn}(\cdot)$ is the sign function, and $(a)_+ = \max\{a, 0\}$). If the minimum eigenvalue of Equation (9.3) solution is greater than δ, this solution is the right step of the generalized gradient descent algorithm. Therefore, to solve Equation (9.2), we can first conduct the soft-threshold and check its minimum eigenvalue. If the solution satisfies the constraint, we take it and continue to the next step; otherwise, we use this solution as the initial value and develop an alternating direction method of multipliers (ADMM) to solve it. In the process of ADMM, one critical step to keeping matrix positive definite is to replace the eigenvalues smaller than δ by δ to match the constraint. This tightened constraint often makes no difference in practice: the problem (9.1) can be solved efficiently.

Remark 1. Tuning matrix Λ largely determines the sparsity of the covariance matrix of residuals. In most instances, the prior information of the structure of the population covariance matrix is non-existent. We now discuss some popular approaches in choosing the tuning parameter. Here, for simplicity, we use the form $\Lambda = \lambda\Psi$, where λ is a constant, Ψ is a $p \times p$ matrix with all off-diagonals take the value of 1, and diagonals take the value of 0. This penalty treats all off-diagonals equally. Henceforth the selection of Λ is the same as choosing the value of λ. The cross-validation method developed by Bien and Tibshirani (2011) to choose the values for λ is:

$$\widehat{\lambda}_{CV} = \arg\max_{\lambda} k^{-1} \sum_{j=1}^{k} \{-\log\det(\widehat{\Sigma}_e(A_j^c)) - \text{tr}(S_{A_j} \widehat{\Sigma}_e^{-1}(A_j^c))\} \tag{9.4}$$

where k is the number of folds that we partition the historical data into e.g., $A_1, A_2, ..., A_k$ subsets, $\hat{\Sigma}_e(A_j^c)$ is our sparse estimator using the complementary set of A_j, and S_{A_j} is the sample covariance matrix using subsets A_j. The pre-setting values of λ are the fine grid of numbers between 0 and a sufficiently large number which can obtain the diagonal matrix. One can also take the (i,j)-th off diagonal element of Λ, $\Lambda_{ij} = 1/\mid S_{ij} \mid$ for off-diagonal elements, which was similar to the adaptive Lasso penalty of Zou (2006). In this paper, we use the Bayesian information criterion (BIC) to choose the value of λ, that is

$$BIC = \log\det(\widehat{\Sigma}_e) + \text{tr}(\widehat{\Sigma}_e^{-1} S_e) + K_{\widehat{\Sigma}} \times \frac{\log(T)}{T} \tag{9.5}$$

where $K_{\hat{\Sigma}}$ is the number of nonzero elements in the sparse covariance matrix estimator, and T is the number of observations. One can use a grid search of λ to maximize the BIC in (9.5).

Remark 2. The values of δ which always concern the minimum eigenvalue of the initial input matrix, are important to get a positive definite $\widehat{\Sigma}_e$. The natural initial input for the algorithm is the sample covariance matrix of residuals S_e. When S_e is not positive definite, such as $p > T$, we can

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firstly transform by the following quadratic optimization problem[8]:

$$S_{e,pd} = \arg\min_{\check{S}_e}\{\frac{1}{2}\|\check{S}_e - S_e\|_F^2 - \tau \log\det(\check{S}_e)\} \tag{9.6}$$

where τ is a small positive number which controls the positive definiteness of $\check{S}_e$. The core idea is using the property of the logarithm barrier function to make the output estimation be positive definite while minimizing the gap between output estimation and the initial target matrix. Problem (9.6) is a convex problem in $\check{S}_e$ so we can easily use quadratic programming to solve it.

9.2 A data-driven cross-validation approach for bandwidth choice

Here we discuss the cross-validation approach to determine the bandwidth for practitioners, which is inspired by the data-driven choice of model parameters in Ding et al. (2021). It is a simple nested cross-validation tool to determine the values of h. Firstly, with data A in hand, we partition it into k chronological order sub-sets $A = (A_1, A_2, \ldots, A_k)$. Then we solve the following objective function to get h:

$$h_{CV} = \arg\max_h \sum_{j=2}^{k}\{SR(\widehat{W}(\cup_{i=1}^{j-1}A_i, h), r_t(A_j))\} \tag{9.7}$$

where k is the number of CV folds, $\widehat{W}(\cup_{i=1}^{j-1}A_i, h)$ is the optimal minimum variance portfolio weight computed using data set $\cup_{i=1}^{j-1}A_i$ and bandwidth h, $r_t(A_j)$ is the excess assets return in data set A_j, which is used as a testing set, and $SR(\widehat{W}, r_t)$ is the realized Sharpe ratio of $\widehat{W}$ by using r_t. In practice, the length of A_1 is $\max\{p/T + 0.1, 0.5\} \times T$, and length of A_j for $j \neq 1$ is $0.1T$.

9.3 Determining the number of factors

It is important to identify the number of factors accurately. Once the number of factors is underestimated, the sparsity of covariance matrix of idiosyncratic residuals would induce more bias, since the true covariance matrix of residuals is not sparse in the true situation. In our setting, the BIC-type information criterion will be used, that is

$$\widehat{m} = \arg\min_m IC(m) \tag{9.8}$$

where $IC(m) = \log V(m, \{\check{B}_x(m)\}) + \frac{p+Th}{pTh}\log(\frac{pTh}{p+Th})m$, and

$$V(m, \{\check{B}_x(m)\}) = \min_{\check{F}=(\check{F}_1,\ldots,\check{F}_T)^\top} \frac{1}{pT}\sum_{i=1}^{p}\sum_{x=1}^{T}[r_{ix} - \check{b}_{ix}(m)^\top \check{F}_x]^2 \tag{9.9}$$

where $\check{B}_x(m) = (pT)^{-1}R^{(x)\top}R^{(x)}\widehat{B}_x(m)$ for $x = 1, 2, ..., T$, and $\widehat{B}_x(m) = (\widehat{b}_{1x}(m), ..., \widehat{b}_{px}(m))^\top$, which is a solution of (3.3) given the number of factor m, and $\check{F}_t$, $\check{F}$ have the same dimension as F_t, F. This is to make sure the sum of squared residuals in (9.9) is robust to any specific estimate of the factor

[8] Alternatively, one could add some small constant ϵI term to make the input sample covariance S_e non-singular, as in Bien and Tibshirani (2011) and Phan et al. (2017). But it might not always give a good solution as shown in their simulation results.

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loadings in (3.3). Under some regularity conditions stated in Su and Wang (2017), it is shown that $\widehat{m}$ is consistent to the true number of factors m_0, that is $P(\widehat{m} = m_0) \to 1$ as $(p, T) \to \infty$.

9.4 Assumptions without the sub-Gaussian condition

As analyzed in Wang et al. (2021), the Assumption 3 can be relaxed into a polynomial rate. We can replace Assumption 3 and Assumption 4 with the following two assumptions, respectively.

Assumption 10. *There exists constants $C_3 > 0$ and $\theta_3 > 2$ such that*

$$\max_{1\le i\le p} \mathbb{E}\left(|e_{it}|^{2\theta_3}\right) \le C_3,$$
$$\max_{1\le j\le m} \mathbb{E}\left(|F_{tj}|^{2\theta_3}\right) \le C_3.$$

Assumption 11. *Let the bandwidth h satisfy $h \asymp T^{-\theta_2}$ and $\frac{T^{1-2\ell}h}{[\log(T)]^3} \to \infty$, where θ_2 and ℓ are two positive constants. In addition, $T^{1/2-\ell(\theta_3-1)}/[\log(T)]^{1/2} = o(h^{1/2})$ and $pT^{1-\ell\theta_3} = o(1)$ with θ_3 defined in Assumption 10.*

Under these two new assumptions, Wang et al. (2021) derived the following result, which has used the sub-Gaussian conditions (i.e. Assumption 3) in previous proof.

Lemma 1. *Suppose that Assumptions 1, 2, 10 and 11 are satisfied. Moreover, $\nu(T, p, h) + h^2 = o(1)$, where $\nu(T, p, h) = \sqrt{\frac{\log(p\vee T)}{Th}}$. Then we have*

$$\max_{1\le i,j\le p} \sup_t \left| \frac{1}{T}\sum_{t=1}^{T} e_{it}e_{jt} - \sigma_{ij}^e \right| = O_p\left(\nu(T, p, h) + h^2\right). \tag{9.10}$$

10 Appendix B: Proofs of Main Theorems

In this section, the proofs of main theorems (Theorem 1, Theorem 3, Theorem 4) are presented.

10.1 Proof of Theorem 1

Proof of Theorem 1. Recall that $\omega(T, p, h) = \sqrt{\frac{\log(T)}{ph} + \frac{\log(p\vee T)}{Th} + h^2}$. Denote

$$\alpha_T = \sqrt{s_1}\omega(T, p, h), \quad \beta_T = \sqrt{p}\omega(T, p, h).$$

Let U be a symmetric matrix of size p, D_U be its diagonal matrix and $R_U = U - D_U$ be its off-diagonal matrix. Then let $\Delta_U = \omega(T, p, h)R_U + \omega(T, p, h)R_U$. Consider the subset $\mathcal{V} = \{U : \|\Delta_U\|_F^2 = C_1^2\alpha_T^2 + C_2^2\beta_T^2\}$. In order to show the rate of convergence for the estimator $\widehat{\Sigma}_e$, it is sufficient to prove

$$P\left(\inf_{U\in\mathcal{V}} L\left(\Sigma_e + \Delta_U\right) > L\left(\Sigma_e\right)\right) \to 1 \tag{10.1}$$

for sufficiently large constants C_1 and C_2. Here $L(\cdot)$ denotes the objective function in (3.4).

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Now we consider the matrix $\Sigma = \Sigma_e + \Delta_U$. The objective function of penalized log-likelihood estimation has the following decomposition

$$L(\Sigma) - L(\Sigma_e) = I_1 + I_2 + I_3,$$

where

$$\begin{aligned} I_1 &= tr(S_e\Omega) + \log|\Sigma| - [tr(S_e\Omega_e) + \log|\Sigma_e|], \\ I_2 &= \sum_{(i,j)\in S^c} \left[\lambda|\sigma_{ij}| - \lambda\left|\sigma_{ij}^e\right|\right], \quad I_3 = \sum_{(i,j)\in S, i\neq j} \left[\lambda|\sigma_{ij}| - \lambda\left|\sigma_{ij}^e\right|\right], \end{aligned}$$

with $\Omega = \Sigma^{-1}$, $\Omega_e = \Sigma_e^{-1}$, $\Sigma_e = \left(\sigma_{ij}^e\right)_{p\times p}$ and $\Sigma = (\sigma_{ij})_{p\times p}$.

Next, we evaluate $I_i, i = 1, 2, 3$. For I_1, it has the following further decomposition

$$I_1 = -tr\left[(S_e - \Sigma_e)\Omega_e\Delta_U\Omega_e\right] + vec(\Delta_U)^\top \left\{\int_0^1 g(v, \Sigma_v)(1-v)dv\right\} vec(\Delta_U) =: K_1 + K_2,$$

where S_{Ω_e} is the sample covariance matrix of a random sample with the population covariance matrix as Ω_e; $\Sigma_v = \Sigma_0 + v\Delta_U$; and

$$g(v, \Sigma_v) = \Sigma_v^{-1} \otimes \Sigma_v^{-1} S_e \Sigma_v^{-1} + \Sigma_v^{-1} S_e \Sigma_v^{-1} \otimes \Sigma_v^{-1} - \Sigma_v^{-1} \otimes \Sigma_v^{-1}.$$

First, let us look at K_1.

$$|K_1| \leq \left|\sum_{(i,j)\in S} (S_{\Omega_e} - \Omega_e)_{ij} (\Delta_U)_{ij}\right| + \left|\sum_{(i,j)\in S^c} (S_{\Omega_e} - \Omega_e)_{ij} (\Delta_U)_{ij}\right| =: J_1 + J_2. \tag{10.2}$$

It follows from Lemma 4 of (Wang et al., 2021), Assumption 1, and the Cauchy-Schwarz inequality that

$$\begin{aligned} J_1 &\leq (s_1 + p)^{1/2} \max_{i,j} \left|(S_{\Omega_e} - \Omega_e)_{ij} (\Delta_U)_{ij}\right| \\ &= O_p\left((s_1 + p)^{1/2}\omega^2(T, p, h)\right) = o_p\left(c_1\alpha_T^2 + c_2\beta_T^2\right). \end{aligned} \tag{10.3}$$

Secondly, we consider $I_2 - J_2$ by taking I_2 and J_2 into account together.

$$\begin{aligned} I_2 - J_2 &\geq \sum_{(i,j)\in S^c} \left[\lambda|\sigma_{ij}| - \tau_2^2\left|(S_{\Omega_e} - \Omega_e)_{ij}\right| \cdot |\sigma_{ij}|\right] \\ &\geq \sum_{(i,j)\in S^c} \left[\lambda|\sigma_{ij}| - \tau_2^2 \max_{i,j}\left|(S_{\Omega_e} - \Omega_e)_{ij}\right| \cdot |\sigma_{ij}|\right] \\ &= \lambda \cdot \sum_{(i,j)\in S^c} \left[1 - \lambda^{-1}W\omega(T, p, h)\right] \cdot |\sigma_{ij}|, \end{aligned}$$

where $W = O_p(1)$ is a positive number satisfying $\tau_2^2 \max_{i,j} |(S_{\Omega_e} - \Omega_e)_{ij}| = W\omega(T, p, h)$. By the

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assumption that $\omega(T,p,h)=o(\lambda)$, we have

$$I_2 - J_2 \geq 0. \tag{10.4}$$

Thirdly, let us evaluate K_2. The term $\|v\Delta_U\Omega_e\|$ is of $o_p(1)$ because

$$\|v\Delta_U\Omega_e\| \leq \|\Delta_U\| \cdot \|\Omega_e\| \leq \tau_2\left(c_1\alpha_T + c_2\beta_T\right) = O\left((\sqrt{p}+\sqrt{s_1})\omega(T,p,h)\right) = o(1),$$

where the last equality uses the condition that $(p+s_1)\omega^2(T,p,h)=o(1)$. Then we apply the Neumann series expansion on Σ_v^{-1} and get

$$\Sigma_v^{-1} = \Omega_e\left(I_p + v\Delta_U\Omega_e\right)^{-1} = \Omega_e\left(I_p - v\Delta_U\Omega_e + o(1)\right), \tag{10.5}$$

where the little o represents a function of the ℓ_1 norm of the residual matrix in the expansion. So $\left\|\Sigma_v^{-1}\right\|$ is evaluated as

$$\left\|\Sigma_v^{-1}\right\| = \|\Omega_e\| + O_p\left(\alpha_T+\beta_T\right) \leq \tau_2 + O_p\left(\alpha_T+\beta_T\right). \tag{10.6}$$

Moreover,

$$\left\|S_e\Sigma_v^{-1}\right\| = \left\|\left(S_e-\Sigma_e\right)\Sigma_v^{-1} + \Sigma_e\Sigma_v^{-1}\right\| = o_p(1) + 1 + O_p\left(\alpha_T+\beta_T\right) = 1 + o_P(1), \tag{10.7}$$

where the second equality uses (10.5), (10.6) and the fact that $\|S_e-\Sigma_e\|_{\max} = O_p(\omega(T,p,h)) = o_P(1)$. Then it follows from (10.5) and (10.7) that

$$g(v,\Sigma_v) = \Omega_e\otimes\Omega_e + O_p(\alpha_T+\beta_T).$$

Hence, the evaluation of K_2 is

$$\begin{aligned} K_2 &= vec(\Delta_U)^\top\left\{\int_0^1 \Omega_e\otimes\Omega_e(1+o_p(1))(1-v)dv\right\}vec(\Delta_U) \\ &\geq \frac{1}{2}\lambda_{\min}(\Omega_e\otimes\Omega_e)\left\|vec(\Delta_U)\right\|^2(1+o_p(1)) \\ &\geq \frac{1}{2}\tau_2^2\left(c_1^2\alpha_T^2 + c_2^2\beta_T^2\right)(1+o_P(1)). \end{aligned} \tag{10.8}$$

Based on the definition of the set $\mathcal{V}$, it is easy to know the order of I_3 as

$$|I_3| = \left|\sum_{(i,j)\in S, i\neq j}\left(|\sigma_{ij}| - \left|\sigma_{ij}^e\right|\right)\right| = O(\alpha_T^2+\beta_T^2). \tag{10.9}$$

Therefore, by (10.3), (10.4), (10.8) and (10.9), we know that K_2 is the leading term of $L(\Sigma)-L(\Sigma_e)$, and then (10.8) indicates that (10.1) is correct. So the proof is completed.

□

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10.2 Proof of Theorem 2

In this part, we utilize the primal-dual witness technique in Ravikumar et al. (2011) to derive the rate of convergence for the estimator $\widehat{\Sigma}_e$ under the ℓ_∞ norm. Before the proof, we introduce some definitions and notations which are used in the proofs in this theorem and the related lemmas.

As mentioned in Ravikumar et al. (2011), the solution to ℓ_1-penalized log-likelihood estimation problem (3.4) could also be treated as a particular type of M-estimator for the population covariance matrix Σ_e, based on minimizing a Bregman divergence between positive definite matrices.

The log-determinant function is of Bregman type because it is strictly convex, continuously differentiable and defined on the closed convex set $\mathcal{S}_+^p$, i.e.

$$\mathcal{S}_+^p := \{A \in R^{p\times p} : A = A^\top, A \geq 0\}.$$

Here $\mathcal{S}_+^p$ consists of all $p \times p$ symmetric positive semi-definite matrices.

For any function g defined on $\mathcal{S}_+^p$ which is of Bregman type, the Bregman divergence is

$$D_g(A||B) = g(A) - g(B) - < \bigtriangledown g(B), A - B > .$$

Consider the log-determinant function defined on $\mathcal{S}_+^p$,

$$g(A) := \begin{cases} -\log\det(A), & A > 0, \\ +\infty, & otherwise. \end{cases}$$

With this definition, the function g is strictly convex and its domain is the set of strictly positive definite matrices. Moreover, it is continuously differentiable over its domain, with

$$\bigtriangledown g(A) = -A^{-1}.$$

The Bregman divergence corresponds to the log-determinant Bregman function g is given by

$$D_g(A||B) = -\log\det(A) + \log\det(B) + tr\left(B^{-1}(A - B)\right). \tag{10.10}$$

So minimizing the Bregman divergence $D_g(\Sigma_e||\Sigma)$ is equivalent to minimizing the objective function

$$tr(\Sigma^{-1}\Sigma_e) + \log\det(\Sigma). \tag{10.11}$$

Since Σ_e is the unknown population covariance matrix, we use the sample covariance matrix S_e to replace it in (10.11) and then derive the first two terms of the objective function in (3.4).

In terms of such equivalence, Ravikumar et al. (2011) show the rate of convergence for the precision matrix estimator under the ℓ_∞ operator norm, by utilizing primal-dual Witness approach. With similar techniques to Theorem 1 of Ravikumar et al. (2011), we will establish the rate of convergence for the covariance matrix estimator $\widehat{\Sigma}_e$ under the factor-augmented model.

Define the The sub-differential of the norm $||\Sigma||_{1,off} := ||\Lambda * \Sigma||_1$ evaluated at some $\Sigma = (\sigma_{ij})_{p\times p}$ is

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the set of all symmetric matrices $Z \in \mathbb{R}^{p\times p}$ such that

$$Z_{ij} = \begin{cases} 0, & i = j; \\ sign(\sigma_{ij}), & i \neq j, \sigma_{ij} \neq 0; \\ \in [-1,1], & i \neq j, \sigma_{ij} = 0. \end{cases} \tag{10.12}$$

Proof of Theorem 2. Let $\mathcal{A}$ denote the event

$$\mathcal{A} := \{||W||_{\max} \leq C\omega(T,p,h)\},$$

where C is a constant. Then $P(\mathcal{A}) \to 1$.

Now we show Lemma 2 is satisfied. Since the regularizer λ is set to be $\frac{8}{\alpha}\omega(T,p,h)$, we have

$$||W||_{\max} \leq \frac{\alpha}{8}\lambda.$$

So it is sufficient to show that

$$||R(\Delta)||_{\max} \leq \frac{\alpha\lambda}{8}.$$

Now we check the assumptions in Lemma 5.

$$2\kappa_{\Gamma^*}\left(||W||_{\max} + \lambda\right) \leq 2C\kappa_{\Gamma^*}\left(1 + \frac{8}{\alpha}\right)\omega(T,p,h) \leq \min\left\{\frac{1}{3\kappa_{\Sigma_e}d}, \frac{1}{3\kappa_{\Sigma_e}^3\kappa_{\Gamma^*}d}\right\}. \tag{10.13}$$

From (11.22) and Lemma 5, we have

$$||\Delta||_{\max} \leq 2\kappa_{\Gamma^*}\left(||W||_{\max} + \lambda\right) \leq 2C\kappa_{\Gamma^*}\left(1 + \frac{8}{\alpha}\right)\omega(T,p,h). \tag{10.14}$$

Then it follows from (10.14) and Lemma 4 that

$$\begin{aligned} ||R(\Delta)||_{\max} &\leq \frac{3}{2}d \cdot ||\Delta||_{\max}^2 \cdot \kappa_{\Sigma_e}^3 \leq 6\kappa_{\Sigma_e}^3\kappa_{\Gamma^*}^2 \cdot d \cdot \left(1 + \frac{8}{\alpha}\right)^2 \cdot \omega^2(T,p,h) \\ &\leq 6\kappa_{\Sigma_e}^3\kappa_{\Gamma^*}^2 \cdot d \cdot \left(1 + \frac{8}{\alpha}\right)^2 \cdot \omega(T,p,h)\frac{\alpha\lambda}{8} \leq \frac{\alpha\lambda}{8}. \end{aligned} \tag{10.15}$$

By Lemma 2, we have

$$\widetilde{\Sigma}_e = \widehat{\Sigma}_e.$$

Then the estimator $\widehat{\Sigma}_e$ satisfies the ℓ_∞ bound (10.14) of $\widetilde{\Sigma}_e$.

Due to the oracle property of the lasso estimation, we have

$$\left|\left|\widehat{\Sigma}_e - \Sigma_e\right|\right|_F = \left[\sum_{i=1}^{p}(\widehat{\sigma}_{ii}^e - \sigma_{ii}^e)^2 + \sum_{(i,j)\in\mathcal{S}}\left(\widehat{\sigma}_{ij}^e - \sigma_{ij}^e\right)^2\right]^{1/2} \leq \sqrt{s_1 + p}\,\omega(T,p,h).$$

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On the other hand, we have

$$\left|\left|\widehat{\Sigma}_e - \Sigma_e\right|\right| \leq \left|\left|\left|\widehat{\Sigma}_e - \Sigma_e\right|\right|\right|_\infty = O_p\left(d\omega(T,p,h)\right).$$

For the precision matrix estimator $\widehat{\Sigma}_e^{-1}$,

$$\begin{aligned} &\left|\left|\widehat{\Sigma}_e^{-1} - \Sigma_e^{-1}\right|\right|_{\max} = \left|\left|\widehat{\Sigma}_e^{-1}\left(\Sigma_e - \widehat{\Sigma}_e\right)\Sigma_e^{-1}\right|\right|_{\max} \\ \leq \quad & ||\Sigma_e||^2 \left|\left|\widehat{\Sigma}_e - \Sigma_e\right|\right|_{\max} = O_p\left(\omega(T,p,h)\right), \end{aligned} \tag{10.16}$$

where the last equality uses Assumption 1 and the assumption that $\left|\left|\widehat{\Sigma}_e - \Sigma_e\right|\right|_{\max} = O_P\left(\omega(T,p,h)\right) = o_P(1)$.

Similarly, we can derive the rates of convergence for the precision matrix estimator $\widehat{\Sigma}_e^{-1}$ under the spectral norm and Frobenius norm, respectively. □

10.3 Proof of Theorem 3

Proof of Theorem 3. Recall that, at any time $t = 1, 2, \ldots, T$,

$$\Sigma_{r,t} = B_t \Sigma_F B_t^\top + \Sigma_e, \quad \widehat{\Sigma}_{r,t} = \widehat{B}_t \widehat{B}_t^\top + \widehat{\Sigma}_e.$$

Define a new matix

$$\widetilde{\Sigma}_{r,t} = B_t S_F B_t^\top + \Sigma_e, \quad t = 1, 2, \ldots, T,$$

where $S_F = \frac{1}{T}\sum_{t=1}^T \widetilde{F}_t \widetilde{F}_t^\top$ and $\widetilde{F}_t = F_t K_{t,h}^{1/2}$.

We evaluate the norm of the term $\widehat{\Sigma}_{r,t} - \Sigma_{r,t}$ in the following way.

$$\sup_t \left|\left|\widehat{\Sigma}_{r,t} - \Sigma_{r,t}\right|\right|_{\Sigma_{r,t}}^2 \leq C \sup_t \left[\left|\left|\widehat{\Sigma}_{r,t} - \widetilde{\Sigma}_{r,t}\right|\right|_{\Sigma_{r,t}} + \left|\left|\widetilde{\Sigma}_{r,t} - \Sigma_{r,t}\right|\right|_{\Sigma_{r,t}}\right]. \tag{10.17}$$

Based on the definition of the norm $\|\cdot\|_{\Sigma_{r,t}}$, for any $p \times p$ matrix A, we have

$$\|A\|_{\Sigma_{r,t}}^2 = \frac{1}{p}\left\|\Sigma_{r,t}^{-1/2} A \Sigma_{r,t}^{-1/2}\right\|_F^2 \leq \frac{C}{p}\|A\|_F^2, \tag{10.18}$$

where C is a constant that may be different from line to line.

First, we evaluate the first term on the right side of (10.17).

$$\sup_t \left|\left|\widehat{\Sigma}_{r,t} - \widetilde{\Sigma}_{r,t}\right|\right|_{\Sigma_{r,t}}^2 \leq C \sup_t \left[\left|\left|\widehat{B}_t \widehat{B}_t^\top - B_t S_F B_t^\top\right|\right|_{\Sigma_{r,t}}^2 + \left|\left|\widehat{\Sigma}_e - \Sigma_e\right|\right|_{\Sigma_{r,t}}^2\right].$$

By (10.18) and Theorem 1, we have

$$\sup_t \left|\left|\widehat{\Sigma}_e - \Sigma_e\right|\right|_{\Sigma_{r,t}}^2 = O_p\left(\sup_t \left|\left|\widehat{\Sigma}_e - \Sigma_e\right|\right|^2\right) = O_p\left(\min\{(p+s_1), d^2\}\omega^2(T,p,h)\right). \tag{10.19}$$

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Secondly, we consider the second term on the right side of (10.17).

$$\sup_t \left\| \widehat{B}_t \widehat{B}_t^\top - B_t S_F B_t^\top \right\|_{\Sigma_{r,t}}^2 \leq C \sup_t \left[\left\| \Delta_{B_t} \Delta_{B_t}^\top \right\|_{\Sigma_{r,t}}^2 + \left\| B H_t^{-1} \Delta_{B_t} \right\|_{\Sigma_{r,t}^\top}^2 + \left\| B_t H_t^{-2} B_t^\top - B_t S_F B_t^\top \right\|_{\Sigma_{r,t}}^2 \right],$$

where $\Delta_{B_t} = \widehat{B}_t - B_t H_t^{-1}$ and $H_t^{-1} = \frac{1}{pT} \left(V_t^{-1} \widehat{F}^\top \widetilde{F} B_t^\top B_t \right)^{-1}$ with V_t being an $m \times m$ diagonal matrix with diagonal elements as the first m largest eigenvalues (in decreasing order) of the matrix $\frac{1}{pT} \widetilde{R} \widetilde{R}^\top$. With the same arguments in (A.17)-(A.20) of Wang et al. (2021), we have

$$\sup_t \left\| \widehat{B}_t \widehat{B}_t^\top - B_t S_F B_t^\top \right\|_{\Sigma_{r,t}}^2 = O_p \left(p\omega^4(T,p,h) + \omega^2(T,p,h) \right) \tag{10.20}$$

It follows from (10.19) and (10.20) that

$$\sup_t \left\| \widehat{\Sigma}_{r,t} - \widetilde{\Sigma}_{r,t} \right\|_{\Sigma_{r,t}}^2 = O_p \left(p\omega^4(T,p,h) + \min\{(p+s_1), d^2\} \omega^2(T,p,h) \right). \tag{10.21}$$

For the term $\widetilde{\Sigma}_{r,t} - \Sigma_{r,t}$, we utilize Assumption 5(2) to obtain

$$\begin{aligned}
& \sup_t \left\| \widetilde{\Sigma}_{r,t} - \Sigma_{r,t} \right\|_{\Sigma_{r,t}}^2 = \sup_t \left\| B_t \left(S_F - \Sigma_F \right) B_t^\top \right\|_{\Sigma_{r,t}}^2 \\
= \; & \frac{1}{p} \sup_t \left\| \Sigma_{r,t}^{-1/2} B_t \left(S_F - \Sigma_F \right) B_t^\top \Sigma_{r,t}^{-1/2} \right\|_F^2 \\
\leq \; & \frac{1}{p} \sup_t \left\| \Sigma_{r,t}^{-1/2} B_t \right\|_F^2 \cdot \left\| S_F - \Sigma_F \right\|^2 = O_p \left(\frac{1}{p} \left(\sqrt{\frac{\log(T)}{Th}} + h^2 \right)^2 \right) \\
= \; & o_P \left(p\omega^4(T,p,h) + \min\{(p+s_1), d^2\} \omega^2(T,p,h) \right).
\end{aligned} \tag{10.22}$$

Therefore, the final evaluation is derived by combining (10.21) and (10.22),

$$\sup_t \left\| \widehat{\Sigma}_{r,t} - \Sigma_{r,t} \right\|_{\Sigma_{r,t}}^2 = O_p \left(p\omega^4(T,p,h) + \min\{(p+s_1), d^2\} \omega^2(T,p,h) \right).$$

Next, we show the order of $\widehat{\Sigma}_{r,t} - \Sigma_{r,t}$ under the ℓ_∞ norm. We decompose this term in the same way as (10.17).

$$\sup_t \left\| \widehat{\Sigma}_{r,t} - \Sigma_{r,t} \right\|_{\max} \leq \sup_t \left\| \widehat{\Sigma}_{r,t} - \widetilde{\Sigma}_{r,t} \right\|_{\max} + \sup_t \left\| \widetilde{\Sigma}_{r,t} - \Sigma_{r,t} \right\|_{\max},$$

where

$$\sup_t \left\| \widehat{\Sigma}_{r,t} - \widetilde{\Sigma}_{r,t} \right\|_{\max} \leq \sup_t \left\| \widehat{\Sigma}_e - \Sigma_e \right\|_{\max} + \sup_t \left\| \widehat{B}_t \widehat{B}_t^\top - B_t \Sigma_F B_t^\top \right\|_{\max}.$$

By Theorem 2, we have

$$\sup_t \left\| \widehat{\Sigma}_e - \Sigma_e \right\|_{\max} = O_p \left(\omega(T,p,h) \right). \tag{10.23}$$

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By the same argument as (A.24) of Wang et al. (2021),

$$\sup_t \left\| \widehat{B}_t \widehat{B}_t^\top - B_t \Sigma_F B_t^\top \right\|_{\max} = O_p\left(\omega(T,p,h)\right). \tag{10.24}$$

With similar proof to (10.22), we have

$$\sup_t \left\| \widetilde{\Sigma}_{r,t} - \Sigma_{r,t} \right\|_{\max} = O_p\left(\sqrt{\frac{\log(T)}{Th}} + h^2\right). \tag{10.25}$$

So it follows from (10.23), (10.24) and (10.25) that

$$\sup_t \left\| \widehat{\Sigma}_{r,t} - \Sigma_{r,t} \right\|_{\max} = O_p\left(\omega(T,p,h)\right). \tag{10.26}$$

Next, we evaluate the rate of convergence for the precision matrix $\widehat{\Sigma}_{r,t}$. Similar to the evaluation for covariance estimation, we utilize the same decomposition by the triangle inequality,

$$\left\| \widehat{\Sigma}_{r,t}^{-1} - \Sigma_{r,t}^{-1} \right\| \leq \left\| \widehat{\Sigma}_{r,t}^{-1} - \widetilde{\Sigma}_{r,t}^{-1} \right\| + \left\| \widetilde{\Sigma}_{r,t}^{-1} - \Sigma_{r,t}^{-1} \right\|.$$

By applying the Sherman-Morrison-Woodbury formula on $\Sigma_{r,t}^{-1}$ and $\widetilde{\Sigma}_{r,t}^{-1}$, we have

$$\begin{aligned}
\Sigma_{r,t}^{-1} &= \Sigma_e^{-1} - \Sigma_e^{-1} B_t \left[\Sigma_F^{-1} + B_t^\top \Sigma_e^{-1} B_t\right]^{-1} B_t^\top \Sigma_e^{-1}, \\
\widetilde{\Sigma}_{r,t}^{-1} &= \Sigma_e^{-1} - \Sigma_e^{-1} B_t \left[S_F^{-1} + B_t^\top \Sigma_e^{-1} B_t\right]^{-1} B_t \Sigma_e^{-1}.
\end{aligned}$$

Then

$$\widetilde{\Sigma}_{r,t}^{-1} - \Sigma_{r,t}^{-1} = -\Sigma_e^{-1} B_t Q B_t^\top \Sigma_e^{-1}, \tag{10.27}$$

where

$$Q = \left[S_F^{-1} + B_t^\top \Sigma_e^{-1} B_t\right]^{-1} - \left[\Sigma_F^{-1} + B_t \Sigma_e^{-1} B_t\right]^{-1} =: Q_1^{-1} - Q_2^{-1}.$$

Since

$$\begin{aligned}
&\left\| Q_1^{-1} - Q_2^{-1} \right\| = \left\| Q_1^{-1} \left(Q_2 - Q_1\right) Q_2^{-1} \right\| \\
\leq\ & \left[\left\| Q_1^{-1} - Q_2^{-1} \right\| + \left\| Q_2 - Q_1 \right\|\right] \left\| Q_2 - Q_1 \right\| \cdot \left\| Q_2^{-1} \right\|,
\end{aligned} \tag{10.28}$$

we have

$$\|Q\| = \left\| Q_1^{-1} - Q_2^{-1} \right\| \leq \frac{\|Q_1 - Q_2\| \cdot \left\| Q_2^{-1} \right\|^2}{1 - \|Q_1 - Q_2\| \cdot \left\| Q_2^{-1} \right\|}. \tag{10.29}$$

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By Assumption 5(2),

$$\sup_t \|Q_1 - Q_2\| = \sup_t \left\|S_F^{-1} - \Sigma_F^{-1}\right\| = O_p\left(\sqrt{\frac{\log(p)}{Th}} + h^2\right). \tag{10.30}$$

Then it follows from (10.27) and (10.30) that

$$\sup_t \left\|\widetilde{\Sigma}_{r,t}^{-1} - \Sigma_{r,t}^{-1}\right\| = O_p\left(\frac{1}{p}\left(\sqrt{\frac{\log(p)}{Th}} + h^2\right)\right) = o_p\left(\min\{\sqrt{p+s_1}, d\} \cdot \omega(T,p,h)\right). \tag{10.31}$$

Next, we consider the term $\left\|\widetilde{\Sigma}_{r,t}^{-1} - \widehat{\Sigma}_{r,t}\right\|$. By applying the Sherman-Morrison-Woodbury formula, we obtain

$$\begin{aligned}
\widehat{\Sigma}_{r,t}^{-1} &= \widehat{\Sigma}_e^{-1} - \widehat{\Sigma}_e^{-1}\widehat{B}_t\left(I_m + \widehat{B}_t^\top \widehat{\Sigma}_e^{-1}\widehat{B}_t\right)^{-1}\widehat{B}_t^\top \widehat{\Sigma}_e^{-1}, \\
\widetilde{\Sigma}_{r,t}^{-1} &= \Sigma_e^{-1} - \Sigma_e^{-1}B_{tH}\left(\Sigma_{FH}^{-1} + B_{tH}^\top \Sigma_e^{-1}B_{tH}\right)^{-1}B_{tH}^\top \Sigma_e^{-1},
\end{aligned}$$

where $B_{tH} = B_t H_t^{-1}$ and $\Sigma_{FH} = H_t \Sigma_F H_t^\top$.

With the same decomposition as proof of Theorem 3 in (Wang et al., 2021), we have

$$\widehat{\Sigma}_{r,t}^{-1} - \widetilde{\Sigma}_{r,t}^{-1} = -\sum_{\ell=1}^{6} D_\ell, \tag{10.32}$$

where

$$\begin{aligned}
D_1 &= \Sigma_e^{-1} - \widehat{\Sigma}_e^{-1}, \\
D_2 &= \left(\widehat{\Sigma}_e^{-1} - \Sigma_e^{-1}\right)\widehat{B}_t\left(I_m + \widehat{B}_t\widehat{\Sigma}_e^{-1}\widehat{B}_t\right)^{-1}\widehat{B}_t^\top \widehat{\Sigma}_e^{-1}, \\
D_3 &= \Sigma_e^{-1}\widehat{B}_t\left(I_m + \widehat{B}^\top \widehat{\Sigma}_e^{-1}\widehat{B}_t\right)^{-1}\widehat{B}_t^\top\left(\widehat{\Sigma}_e^{-1} - \Sigma_e^{-1}\right), \\
D_4 &= \Sigma_e^{-1}\left(\widehat{B}_t - B_t\right)\left(I_m + \widehat{B}_t^\top \widehat{\Sigma}_e^{-1}\widehat{B}_t\right)^{-1}\widehat{B}_t^\top \Sigma_e^{-1}, \\
D_5 &= \Sigma_e^{-1}B_{tH}\left(I_m + \widehat{B}_t\widehat{\Sigma}_e^{-1}\widehat{B}_t\right)^{-1}\left(\widehat{B}_t - B_{tH}\right)^\top \Sigma_e^{-1}, \\
D_6 &= \Sigma_e^{-1}B_{tH}\widetilde{Q}_H B_{tH}^\top \Sigma_e^{-1},
\end{aligned}$$

with

$$\widetilde{Q} = \left(I_m + \widehat{B}_t^\top \widehat{\Sigma}_e^{-1}\widehat{B}_t\right)^{-1} - \left(\Sigma_{FH}^{-1} + B_{tH}^\top \Sigma_e^{-1}B_{tH}\right)^{-1} =: \widetilde{Q}_1^{-1} - \widetilde{Q}_2^{-1}.$$

By Theorem 1, we have

$$\sup_t \|D_1\| = \sup_t \left\|\widehat{\Sigma}_e^{-1} - \Sigma_e^{-1}\right\| = O_p\left(\min\{\sqrt{p+s_1}, d\} \cdot \omega(T,p,h)\right). \tag{10.33}$$

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For D_2,

$$\begin{aligned}\sup_t \|D_2\| &\leq \sup_t \left\|\widehat{\Sigma}_e^{-1} - \Sigma_e^{-1}\right\| \cdot \left\|\widehat{B}_t \left(I_m + \widehat{B}_t^\top \widehat{\Sigma}_e^{-1} \widehat{B}_t\right)^{-1} \widehat{B}_t^\top\right\|^{-1} \left\|\widehat{\Sigma}_e^{-1}\right\| \\ &= O_p\left(\min\{\sqrt{p+s_1}, d\} \cdot \omega(T,p,h)\right), \end{aligned} \tag{10.34}$$

where the last equality has used the fact that

$$\sup_t \left\|\left(I_m + \widehat{B}_t \widehat{\Sigma}_e^{-1} \widehat{B}_t^\top\right)^{-1}\right\| = O_p\left(\frac{1}{p}\right).$$

The order of $\sup_t \|D_3\|$ can be derived in a similar way.

$$\sup_t \|D_3\| = O_p\left(\min\{\sqrt{p+s_1}, d\} \cdot \omega(T,p,h)\right). \tag{10.35}$$

For D_4,

$$\begin{aligned}\sup_t \|D_4\| &\leq \sup_t \left\|\Sigma_e^{-1}\left(\widehat{B}_t - B_{tH}\right)\right\| \cdot \left\|I_m + \widehat{B}_t^\top \widehat{\Sigma}_e^{-1} \widehat{B}_t\right\|^{-1} \left\|\widehat{B}_t \widehat{\Sigma}_e^{-1}\right\| \\ &= O_p\left(\omega(T,p,h)\right), \end{aligned} \tag{10.36}$$

where the last equality uses (S21) of (Wang et al., 2021), i.e.

$$\max_{1\leq i\leq p} \sup_t \left\|\widehat{B}_t - \left(H_t^{-1}\right)^\top B_t\right\|^2 = O_p\left(\omega^2(T,p,h)\right).$$

The order of $\|D_5\|$ is obtained similarly, i.e.

$$\sup_t \|D_5\| = O_p\left(\omega(T,p,h)\right). \tag{10.37}$$

For $\widetilde{Q}$, by using (10.29),

$$\sup_t \left\|\widetilde{Q}\right\| = \sup_t \left\|\widetilde{Q}_1^{-1} - \widetilde{Q}_2^{-1}\right\| \leq \frac{\left\|\widetilde{Q}_1 - \widetilde{Q}_2\right\| \cdot \left\|\widetilde{Q}_2^{-1}\right\|^2}{1 - \left\|\widetilde{Q}_1 - \widetilde{Q}_2\right\| \cdot \left\|\widetilde{Q}_2^{-1}\right\|} \tag{10.38}$$

It follows from (S15), (S21) of (Wang et al., 2021) and Theorem 1 that

$$\sup_t \left\|\widetilde{Q}\right\| = O_p\left(\frac{\min\{\sqrt{p+s_1}, d\}}{p} \omega(T,p,h)\right).$$

So

$$\sup_t \|D_6\| = O_p\left(\min\{\sqrt{p+s_1}, d\} \cdot \omega(T,p,h)\right). \tag{10.39}$$

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By (10.33)-(10.39), we obtain

$$\sup_t \left\| \widehat{\Sigma}_{r,t}^{-1} - \widetilde{\Sigma}_{r,t}^{-1} \right\| = O_p \left(\min\{\sqrt{p+s_1}, d\} \cdot \omega(T,p,h) \right). \tag{10.40}$$

The final result is derived from (10.31) and (10.40). □

10.4 Proof of Theorem 4

Proof of Theorem 4. For convenient reference, we recall some notations for essential estimators and criteria as follows.

$$\begin{aligned}
&\widehat{W}_t = \frac{\widehat{\Sigma}_{r,t}^{-1} 1_p}{1_p^\top \widehat{\Sigma}_{r,t}^{-1} 1_p}, \quad W_t = \frac{\Sigma_{r,t}^{-1} 1_p}{1_p^\top \Sigma_{r,t}^{-1} 1_p}, \\
&\widehat{R}_{\min,t} = \frac{1}{1_p^\top \widehat{\Sigma}_{r,t}^{-1} 1_p}, \quad R_{\min,t} = \frac{1}{1_p^\top \Sigma_{r,t}^{-1} 1_p}, \\
&\widehat{R}\left(W_t\right) = W_t^\top \widehat{\Sigma}_{r,t}^{-1} W_t, \quad R\left(W_t\right) = W_t^\top \Sigma_{r,t}^{-1} W_t.
\end{aligned}$$

First, we evaluate the estimator $\widehat{R}_{\min,t}$.

$$\left| \frac{\widehat{R}_{\min,t}}{R_{\min,t}} - 1 \right| = \left| \frac{1_p^\top \Sigma_{r,t}^{-1} 1_p}{1_p^\top \widehat{\Sigma}_{r,t}^{-1} 1_p} - 1 \right| = \frac{\left| 1_p^\top \Sigma_{r,t}^{-1} 1_p - 1_p^\top \widehat{\Sigma}_{r,t}^{-1} 1_p \right|}{1_p^\top \widehat{\Sigma}_{r,t}^{-1} 1_p} \leq \frac{p \cdot \left\| \Sigma_{r,t}^{-1} - \widehat{\Sigma}_{r,t}^{-1} \right\|}{1_p^\top \widehat{\Sigma}_{r,t}^{-1} 1_p}. \tag{10.41}$$

The bound for the numerator of (10.41) is $O_p\left(p \min\{\sqrt{p+s_1}, d\}\omega(T,p,h)\right)$, which is derived in Theorem 3. The exact order of the denominator of (10.41) is $p^{1-\eta}$, which is guaranteed by Assumption 8 and the same argument as the Proposition of Ding et al. (2021). In view of these results, we have the result

$$\left| \frac{\widehat{R}_{\min,t}}{R_{\min,t}} - 1 \right| = O_p \left(p^\eta \min\{\sqrt{p+s_1}, d\}\omega(T,p,h) \right). \tag{10.42}$$

Now let us look at the estimator $R\left(\widehat{W}_t\right)$.

$$\begin{aligned}
\frac{R\left(\widehat{W}_t\right)}{R(W_t)} &= \frac{\widehat{W}_t^\top \Sigma_{r,t} \widehat{W}_t}{W_t^{*\top} \Sigma_{r,t} W_t^*} = \left(\frac{1_p^\top \widehat{\Sigma}_{r,t}^{-1} \Sigma_{r,t} \widehat{\Sigma}_{r,t}^{-1} 1_p}{1_p^\top \Sigma_{r,t}^{-1} 1_p} \right) \left(\frac{1_p^\top \Sigma_{r,t}^{-1} 1_p}{1_p^\top \widehat{\Sigma}_{r,t}^{-1} 1_p} \right)^2 \\
&= \left(\frac{1_p^\top \left(\widehat{\Sigma}_{r,t}^{-1} - \Sigma_{r,t}^{-1} \right) \Sigma_{r,t} \widehat{\Sigma}_{r,t}^{-1} 1_p}{1_p^\top \Sigma_{r,t}^{-1} 1_p} \right) \left(\frac{1_p^\top \Sigma_{r,t}^{-1} 1_p}{1_p^\top \widehat{\Sigma}_{r,t}^{-1} 1_p} \right)^2 + \frac{1_p^\top \Sigma_{r,t}^{-1} 1_p}{1_p^\top \widehat{\Sigma}_{r,t}^{-1} 1_p} \\
&=: A_1 \times A_2^2 + A_2.
\end{aligned}$$

In terms of (10.42), the order of A_2 is

$$A_2 = 1 + O_p \left(p^\eta \min\{\sqrt{p+s_1}, d\}\omega(T,p,h) \right). \tag{10.43}$$

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Next, we show that

$$A_1 = O_p\left(p^{\eta}\min\{\sqrt{p+s_1}, d\}\omega(T,p,h)\right). \tag{10.44}$$

The term $|A_1|$ could be evaluated in the following way.

$$\begin{aligned}
|A_1| &\leq \frac{p\left\|\widehat{\Sigma}_{r,t}^{-1} - \Sigma_{r,t}^{-1}\right\|}{1_p^\top \Sigma_{r,t}^{-1} 1_p}\left\|\Sigma_{r,t}\widehat{\Sigma}_{r,t}^{-1}\right\| \\
&= O_p\left(p^{\eta}\min\{\sqrt{p+s_1}, d\}\omega(T,p,h)\right)\cdot\left\|\widehat{\Sigma}_{r,t}^{-1/2}\Sigma_{r,t}\widehat{\Sigma}^{-1/2}\right\|.
\end{aligned}$$

By the decomposition

$$\widehat{\Sigma}_{r,t}^{-1/2}\Sigma_{r,t}\widehat{\Sigma}_{r,t}^{-1/2} = \widehat{\Sigma}_{r,t}^{-1/2}\left(\Sigma_{r,t} - \widehat{\Sigma}_{r,t}\right)\widehat{\Sigma}_{r,t}^{-1/2} + I_p,$$

we have

$$\begin{aligned}
\left\|\widehat{\Sigma}_{r,t}^{-1/2}\Sigma_{r,t}\widehat{\Sigma}_{r,t}^{-1/2}\right\| - 1 &\leq \left\|\widehat{\Sigma}^{-1/2}\left(\widehat{B}_t\widehat{B}_t^\top - B_tB_t^\top\right)\widehat{\Sigma}_{r,t}^{-1/2}\right\| + \left\|\widehat{\Sigma}_{r,t}^{-1/2}\left(\widehat{\Sigma}_e - \Sigma_e\right)\widehat{\Sigma}_{r,t}^{-1/2}\right\| \\
&\leq \left\|\widehat{\Sigma}_{r,t}^{-1/2}\left(\widehat{B}_tH_t^{-1\top} - B_t\right)\left(\widehat{B}_tH_t^{-1\top} - B_t\right)^\top\widehat{\Sigma}_{r,t}^{-1/2}\right\| \\
&\quad +2\left\|\widehat{\Sigma}^{-1/2}\left(\widehat{B}H_t^{-1\top} - B_t\right)H_t^{-1}\widehat{B}_t^\top\widehat{\Sigma}_{r,t}^{-1/2}\right\| \\
&\quad +\left\|\widehat{\Sigma}_{r,t}^{-1/2}\widehat{B}_t\left(I_p - \left(H_tH_t^\top\right)^{-1}\right)\widehat{B}_t^\top\widehat{\Sigma}_{r,t}^{-1/2}\right\| \\
&\quad +\left\|\widehat{\Sigma}_{r,t}^{-1/2}\left(\widehat{\Sigma}_e - \Sigma_e\right)\widehat{\Sigma}^{-1/2}\right\| \\
&=: B_1 + B_2 + B_3 + B_4.
\end{aligned}$$

Then we evaluate these four terms $B_i, i = 1,2,3,4$. For the term B_1, we have

$$\begin{aligned}
B_1 &= \left\|\widehat{\Sigma}_{r,t}^{-1/2}\left(\widehat{B}_t - B_tH_t^\top\right)\left(H_tH_t^\top\right)^{-1}\left(\widehat{B}_t - B_tH_t^\top\right)^\top\widehat{\Sigma}_{r,t}^{-1/2}\right\| \\
&\leq \left\|\widehat{\Sigma}_{r,t}^{-1/2}\right\|^2\left\|\widehat{B}_t - B_tH_t^\top\right\|^2\left\|\left(H_tH_t^\top\right)^{-1}\right\|^2 \\
&= O_p\left(\sqrt{p}\omega(T,p,h)\right),
\end{aligned}$$

where the last equality uses (S.21) of Wang et al. (2021).

Similarly, B_2 is estimated below.

$$\begin{aligned}
B_2 &= 2\left\|\widehat{\Sigma}_t^{-1/2}\left(\widehat{B}_t - B_tH_t^\top\right)\left(H_tH_t^\top\right)^{-1}\widehat{B}_t^\top\widehat{\Sigma}_t^{-1/2}\right\| \\
&= O_p\left(p^{1/4}\omega(T,p,h)\right),
\end{aligned}$$

where the last equality follows the same argument of B_1 above.

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For the term B_3,

$$B_3 \leq \left\| \widehat{B}_t^\top \widehat{\Sigma}_{r,t}^{-1/2} \right\|^2 \cdot \left\| I_p - \left(H_t H_t^\top \right)^{-1} \right\| = O_p \left(\omega(T,p,h) \right),$$

where the last equality uses (S.19) of Wang et al. (2021).

It follows from Theorem 1 that

$$B_4 \leq \left\| \widehat{\Sigma}_{r,t}^{-1} \right\| \cdot \left\| \widehat{\Sigma}_e - \Sigma_e \right\| = O_p \left(\min\{\sqrt{p+s_1}, d\} \omega(T,p,h) \right).$$

Combining these evaluations for $B_i, i = 1,2,3,4$, we have

$$\left\| \widehat{\Sigma}_{r,t}^{-1/2} \Sigma_{r,t} \widehat{\Sigma}_{r,t}^{-1/2} \right\| - 1 = O_p \left(p^{1/4} \omega(T,p,h) + \min\{\sqrt{p+s_1}, d\} \omega(T,p,h) \right). \tag{10.45}$$

By Assumption 8 and (10.45), the result (10.44) is derived. □

10.5 Proof of Theorem 5

Proof. We decompose the Sharpe ratio in the following way.

$$\frac{\widehat{SR}_t - SR_t}{SR_t} = \frac{A_1 + A_2}{\sqrt{1^\top \widehat{\Sigma}_{r,t}^{-1} 1} \cdot 1^\top \Sigma_{r,t}^{-1} \mu_t},$$

where

$$A_1 = 1^\top \left(\widehat{\Sigma}_{r,t}^{-1} \widehat{\mu}_t - \Sigma_{r,t}^{-1} \mu_t \right) \cdot \sqrt{1^\top \Sigma_{r,t}^{-1} 1},$$
$$A_2 = 1^\top \Sigma_{r,t}^{-1} \mu_t \left(\sqrt{1^\top \Sigma_{r,t}^{-1} 1} - \sqrt{1^\top \widehat{\Sigma}_{r,t}^{-1} 1} \right).$$

Now we consider the first term involving A_1,

$$\begin{aligned}
||B_1|| &:= \frac{||A_1||}{\sqrt{1^\top \widehat{\Sigma}_{r,t}^{-1} 1} \cdot \left| 1^\top \Sigma_{r,t}^{-1} \mu_t \right|} \\
&\leq \frac{\left| 1^\top \left(\widehat{\Sigma}_{r,t}^{-1} - \Sigma_{r,t}^{-1} \right) \widehat{\mu}_t \right| + \left| 1^\top \Sigma_{r,t}^{-1} \left(\widehat{\mu}_t - \mu_t \right) \right|}{\sqrt{1^\top \widehat{\Sigma}_{r,t}^{-1} 1} \cdot \left| 1^\top \Sigma_{r,t}^{-1} \mu_t \right|} \cdot \sqrt{1^\top \Sigma_{r,t}^{-1} 1} \\
&\leq M \cdot \left(p^\delta \left\| \widehat{\Sigma}_{r,t}^{-1} - \Sigma_{r,t}^{-1} \right\| + p^\delta ||\widehat{\mu}_t - \mu_t|| \right) \\
&= O_p \left(p^\delta \min\{\sqrt{s_1 + p}, d\} \cdot \omega(T,p,h) + p^\delta \sqrt{p} \omega(T,p,h) \right),
\end{aligned} \tag{10.46}$$

where the last equality uses (S.21) of Wang et al. (2021) which results in

$$\sup_t ||\widehat{\mu}_t - \mu_t|| = O_p \left(\sqrt{p} \omega(T,p,h) \right).$$

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Similarly, we can get

$$||B_2|| := \frac{||A_2||}{\sqrt{1^\top \widehat{\Sigma}_{r,t}^{-1} 1} \cdot \left|1^\top \Sigma_{r,t}^{-1} \mu_t\right|} \leq \frac{\left|\sqrt{1^\top \Sigma_{r,t}^{-1} 1} - \sqrt{1^\top \widehat{\Sigma}_{r,t}^{-1} 1}\right|}{\sqrt{1^\top \widehat{\Sigma}_{r,t}^{-1} 1}} \tag{10.47}$$

$$= \frac{\left|1^\top \left(\Sigma_{r,t}^{-1} - \widehat{\Sigma}_{r,t}^{-1}\right) 1\right|}{\sqrt{1^\top \widehat{\Sigma}_{r,t}^{-1} 1} \left(\sqrt{1^\top \Sigma_{r,t}^{-1} 1} + \sqrt{1^\top \widehat{\Sigma}_{r,t}^{-1} 1}\right)} \leq M \left|\left|\widehat{\Sigma}_{r,t}^{-1} - \Sigma_{r,t}^{-1}\right|\right|$$

$$= O_p\left(\min\{\sqrt{s_1 + p}, d\}\omega(T, p, h)\right). \tag{10.48}$$

Then the result is derived from (10.46) and (10.47). □

11 Appendix C: Lemmas related to Theorem 2

In this section, we will introduce some lemmas used in proofs of Theorem 2.

Lemma 2 (First Derivative Equation)**.** *For any $\lambda > 0$ and sample covariance matrix S_e with strictly positive diagonal elements, the ℓ_1-regularized* log*-determined problem has a local minimizer solution $\widehat{\Sigma}_e \succ 0$ satisfying*

$$-\widehat{\Sigma}_e^{-1} S_e \widehat{\Sigma}_e^{-1} + \widehat{\Sigma}_e^{-1} + \lambda \widehat{Z} = 0, \tag{11.1}$$

where $\widehat{Z}$ is an element of the sub-differential $\partial||\widehat{\Sigma}_e||_{1,off}$.

The procedure of primal-dual witness approach is illustrated as follows.

1. Consider a new matrix $\widetilde{\Sigma}_e$ which is a solution to the restricted log-determinant problem, i.e.

$$\widetilde{\Sigma}_e := \arg \min_{\Sigma \succ 0, \Sigma = \Sigma^\top, \Sigma_{S^c} = 0} \left[tr\left(\Sigma^{-1} S_e\right) + \log\det(\Sigma) + \lambda \cdot ||\Sigma||_{1,off}\right]. \tag{11.2}$$

   Based on this construction, we know that the restricted estimator $\widetilde{\Sigma}_e \succ 0$ and $\widetilde{\Sigma}_{e,\mathcal{S}^c} = 0$.

2. Define $\widetilde{Z}$ as a member of the sub-differential of the regularizer $||\cdot||_{1,off}$ at the point $\widetilde{\Sigma}_e$.

3. To make the restricted estimator $\widetilde{Z}$ satisfy the condition (11.1), we change the elements of $\widetilde{Z}$ on the subset $\mathcal{S}^c$ into the following form

$$\widetilde{Z}_{ij} := \frac{1}{\lambda_n}\left[\left(\widetilde{\Sigma}_e^{-1} S_e \widetilde{\Sigma}_e^{-1}\right)_{ij} - \left[\widetilde{\Sigma}_e^{-1}\right]_{ij}\right], \quad \forall (i,j) \in \mathcal{S}^c. \tag{11.3}$$

4. We will verify the "strict dual feasibility" condition below

$$\left|\widetilde{Z}_{ij}\right| < 1, \quad \forall (i,j) \in \mathcal{S}^c.$$

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We will show that the primal-dual witness technique succeeds in probability. Then we can conclude that the support of the optimal solution $\widehat{\Sigma}_e$ is contained within the support of the true matrix Σ_e. Furthermore, we provide the elementwise ℓ_∞ bound for the difference $\widehat{\Sigma}_e - \Sigma_e$, i.e. $||\widehat{\Sigma}_e - \Sigma_e||_{\max}$

Before starting the proof, we specify some notations first.

$$W := S_e - \Sigma_e, \quad \Delta := \widetilde{\Sigma}_e - \Sigma_e, \quad R(\Delta) := \widetilde{\Sigma}_e^{-1} - \Sigma_e^{-1} + \Sigma_e^{-1}\Delta\Sigma_e^{-1}.$$

By the definition of $\widetilde{\Sigma}_e$ in (11.2), we have $\Delta_{\mathcal{S}^c} = 0$. The notation $R(\Delta)$ is the difference of the gradient $\bigtriangledown g(\widetilde{\Sigma}_e) = \widetilde{\Sigma}_e^{-1}$ from its first-order Taylor expansion around the true population covariance matrix Σ_e.

Next, we provide several auxiliary results.

Lemma 3 (Strict Dual Feasibility)**.** *Assume that*

$$\max\{||W||_{\max}, ||\Delta||_{\max}, ||R(\Delta)||_{\max}\} \leq \frac{\alpha\lambda}{8}. \tag{11.4}$$

Then the vector $\widetilde{Z}_{\mathcal{S}^c}$ satisfies

$$\left|\left|\widetilde{Z}_{\mathcal{S}^c}\right|\right| < 1, \quad \textit{and then} \quad \widetilde{\Sigma}_e = \widehat{\Sigma}_e.$$

Proof. In terms of (11.2) and (11.3), we have

$$-\widetilde{\Sigma}_e^{-1} S_e \widetilde{\Sigma}_e^{-1} + \widetilde{\Sigma}_e + \lambda\widetilde{Z} = 0. \tag{11.5}$$

Based on the definitions of $R(\Delta)$ and Δ, the Equation (11.5) can be represented in the following form.

$$\Sigma_e^{-1}\Delta\Sigma_e^{-1} + V - R(\Delta) - \lambda\widetilde{Z} = 0, \tag{11.6}$$

where $V = \widetilde{\Sigma}_e^{-1} S_e \widetilde{\Sigma}_e^{-1} - \Sigma_e^{-1}$. For convenient demonstration, we represent (11.6) in the vector form. Since the vector form of the matrix $\Sigma_e^{-1}\Delta\Sigma_e^{-1}$ is

$$vec\left(\Sigma_e^{-1}\Delta\Sigma_e^{-1}\right) = \left(\Sigma_e^{-1} \otimes \Sigma_e^{-1}\right) \cdot vec(\Delta) =: \Gamma \cdot vec(\Delta),$$

and the fact that $vec(\Delta)_{\mathcal{S}^c} = 0$, we get the equivalent forms to (11.6) as

$$\Gamma^*_{\mathcal{SS}} \cdot vec(\Delta)_{\mathcal{S}} + vec(V)_{\mathcal{S}} - vec(R(\Delta))_{\mathcal{S}} - \lambda \cdot vec(\widetilde{Z})_{\mathcal{S}} = 0, \tag{11.7}$$

$$\Gamma^*_{\mathcal{S}^c\mathcal{S}} vec(\Delta)_{\mathcal{S}} + vec(V)_{\mathcal{S}^c} - vec(R(\Delta))_{\mathcal{S}^c} - \lambda \cdot vec(\widetilde{Z})_{\mathcal{S}^c} = 0. \tag{11.8}$$

It follows from (11.7) that

$$vec(\Delta)_{\mathcal{S}} = (\Gamma^*_{\mathcal{SS}})^{-1} \cdot \left[-vec(V)_{\mathcal{S}} + vec(R(\Delta))_{\mathcal{S}} + \lambda \cdot vec(\widetilde{Z})_{\mathcal{S}}\right]. \tag{11.9}$$

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By substituting $vec(\Delta)_S$ in (11.8) with the expression (11.9), we have

$$\begin{aligned} vec(\widetilde{Z})_{S^c} &= \frac{1}{\lambda}\Gamma^*_{S^cS}\left(\Gamma^*_{SS}\right)^{-1}\cdot\left[-vec(V)_S + vec(R(\Delta))_S + \lambda\cdot vec(\widetilde{Z})_S\right] \\ &\quad +\frac{1}{\lambda}vec(V)_{S^c} - \frac{1}{\lambda}\cdot vec(R(\Delta))_{S^c} \\ &= \frac{1}{\lambda}\Gamma^*_{S^cS}\left(\Gamma^*_{SS}\right)^{-1}\left(vec(R(\Delta)) - vec(V)_S\right) \\ &\quad +\Gamma^*_{S^cS}\left(\Gamma^*_{SS}\right)^{-1} vec(\widetilde{Z})_S \\ &\quad +\frac{1}{\lambda}\left[vec(V)_S - vec(R(\Delta))_{S^c}\right]. \end{aligned}$$

Then

$$\begin{aligned} &\left|\left|vec(\widetilde{Z})_{S^c}\right|\right| \leq \frac{1}{\lambda}\cdot\left|\left|\left|\Gamma^*_{S^cS}\left(\Gamma^*_{SS}\right)^{-1}\right|\right|\right|_{\max}\cdot\left(||vec(R(\Delta))_S||_{\max} + ||vec(V)_S||_{\max}\right) \\ &+\left|\left|\Gamma^*_{S^cS}\left(\Gamma^*_{SS}\right)^{-1} vec(\widetilde{Z})_S\right|\right|_{\max} + \frac{1}{\lambda}\left(||vec(V)_{S^c}||_{\max} + ||vec(R(\Delta))_{S^c}||_{\max}\right). \end{aligned} \tag{11.10}$$

Based on Assumption 7, it follows that

$$\left|\left|\Gamma^*_{S^cS}\left(\Gamma^*_{SS}\right)^{-1} vec\left(\widetilde{Z}_{S^c}\right)\right|\right|_{\max} \leq \left|\left|\Gamma^*_{S^cS}\left(\Gamma^*_{SS}\right)^{-1}\right|\right|_1 \leq 1-\alpha. \tag{11.11}$$

Then, from (11.10) and (11.11), we have

$$\begin{aligned} \left|\left|vec\left(\widetilde{Z}\right)_{S^c}\right|\right|_{\max} &\leq \frac{2}{\lambda}\left(||vec(V)||_{\max} + ||vec(R(\Delta))||_{\max}\right) \\ &\leq \frac{2-\alpha}{\lambda}\cdot\frac{\alpha\lambda}{8} + 1 - \alpha \leq \frac{\alpha}{2} + 1 - \alpha < 1. \end{aligned}$$

□

Lemma 4 (Control of the Remainder $R(\Delta)$)**.** *Suppose that*

$$||\Delta||_{\max} \leq \frac{1}{3d\cdot\left|\left|\left|\Sigma_e^{-1}\right|\right|\right|_\infty}. \tag{11.12}$$

Then the matrix $J := \sum_{k=0}^{\infty}(-1)^k\left(\Sigma_e^{-1}\Delta\right)^k$ *satisfies*

$$|||J|||_\infty \leq \frac{3}{2} \tag{11.13}$$

and the matrix $R(\Delta) = \Sigma_e^{-1}\Delta\Sigma_e^{-1}\Delta J\Sigma_e^{-1}$ *satisfies*

$$||R(\Delta)||_{\max} \leq \frac{3}{2}d\cdot||\Delta||^2_{\max}\cdot\left|\left|\left|\Sigma_e^{-1}\right|\right|\right|^3_\infty, \tag{11.14}$$

where d *is the maximal number of nonzero elements per row of* Σ_e*.*

Proof. Recall that

$$R(\Delta) = (\Sigma_e + \Delta)^{-1} - \Sigma_e^{-1} + \Sigma_e^{-1}\Delta\Sigma_e^{-1}.$$

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Now we evaluate $||R(\Delta)||_{\max}$. It follows from the condition (11.12) that

$$\left|\left|\left|\Sigma_e^{-1}\Delta\right|\right|\right|_\infty \leq \left|\left|\left|\Sigma_e^{-1}\right|\right|\right|_\infty \cdot |||\Delta|||_\infty \leq \left|\left|\left|\Sigma_e^{-1}\right|\right|\right|_\infty \cdot d\,||\Delta||_{\max} < \frac{1}{3}.$$

Consider the convergent matrix expansion

$$\begin{aligned}
(\Sigma_e+\Delta)^{-1} &= \left[\Sigma_e\left(I_p+\Sigma_e^{-1}\Delta\right)\right]^{-1} = \left(I_p+\Sigma_e^{-1}\Delta\right)^{-1}\Sigma_e^{-1} = \sum_{k=0}^{\infty}(-1)^k\left(\Sigma_e^{-1}\Delta\right)^k\Sigma_e^{-1} \\
&= \Sigma_e^{-1}-\Sigma_e^{-1}\Delta\Sigma_e^{-1}+\sum_{k=2}^{\infty}(-1)^k\left(\Sigma_e^{-1}\Delta\right)^k\Sigma_e^{-1} \\
&= \Sigma_e^{-1}-\Sigma_e^{-1}\Delta\Sigma_e^{-1}+\Sigma_e^{-1}\Delta\Sigma_e^{-1}\Delta J\Sigma_e^{-1},
\end{aligned}$$

where J is defined above (11.13).

Then

$$\begin{aligned}
||R(\Delta)||_{\max} &= \max_{i,j}\left|o_i^\top\Sigma_e^{-1}\Delta\Sigma_e^{-1}\Delta J\Sigma_e^{-1}o_j\right| \\
&\leq \max_i\left|\left|o_i^\top\Sigma_e^{-1}\Delta\right|\right|_{\max}\cdot\max_j\left|\left|\Sigma_e^{-1}\Delta J\Sigma_e^{-1}o_j\right|\right|_1 \\
&\leq \max_i\left|\left|o_i^\top\Sigma_e^{-1}\right|\right|_1\cdot||\Delta||_{\max}\cdot\max_j\left|\left|\Sigma_e^{-1}\Delta J\Sigma_e^{-1}o_j\right|\right|_1, \\
&\leq \left|\left|\left|\Sigma_e^{-1}\right|\right|\right|_\infty\cdot||\Delta||_{\max}\cdot\left|\left|\left|\Sigma_e^{-1}\Delta J\Sigma_e^{-1}\right|\right|\right|_1 \\
&= \left|\left|\left|\Sigma_e^{-1}\right|\right|\right|_\infty\cdot||\Delta||_{\max}\cdot\left|\left|\left|\Sigma_e^{-1}\Delta J\Sigma_e^{-1}\right|\right|\right|_\infty \\
&\leq \left|\left|\left|\Sigma_e^{-1}\right|\right|\right|_\infty\cdot||\Delta||_{\max}\cdot\left|\left|\left|\Sigma_e^{-1}\right|\right|\right|_\infty^2\cdot|||J|||_\infty\cdot|||\Delta|||_\infty, \qquad (11.15)
\end{aligned}$$

where the first two inequalities use the facts that, for any $a, b \in \mathbb{R}^p$,

$$\left|a^\top b\right| \leq ||a||_{\max}\cdot||b||_1, \quad \left|\left|a^\top\Delta\right|\right|_{\max} \leq ||a||_1\cdot||\Delta||_{\max},$$

and the last the equality uses the fact that, for any matrix A,

$$|||A|||_1 = |||A|||_\infty.$$

Because

$$|||J|||_\infty \leq \sum_{k=0}^{\infty}\left|\left|\left|\Delta\Sigma_e^{-1}\right|\right|\right|_\infty^k \leq \frac{1}{1-|||\Sigma_e^{-1}|||_\infty\cdot|||\Delta|||_\infty} \leq \frac{3}{2}, \qquad (11.16)$$

together with (11.15), we obtain

$$||R(\Delta)||_{\max} \leq \frac{3}{2}||\Delta||_{\max}\cdot|||\Sigma_e^{-1}|||_\infty\cdot|||\Sigma_e^{-1}|||_\infty^2\cdot|||\Delta|||_\infty^3, \qquad (11.17)$$

where the last inequality uses $|||\Delta|||_\infty \leq d||\Delta||_{\max}$.

□

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Lemma 5 (Control of Δ)**.** *Suppose that*

$$L := 2\kappa_{\Gamma^*}(||W||_{\max} + \lambda) \leq \min\left\{\frac{1}{3\kappa_{\Sigma_e} d}, \frac{1}{3\kappa_{\Sigma_e}^3 \kappa_{\Gamma^*} d}\right\}$$

Then we have

$$||\Delta||_{\max} = \left|\left|\widetilde{\Sigma}_e - \Sigma_e\right|\right|_{\max} \leq L.$$

Proof. We define a continuous map $F : \Delta_S \to F(\Delta_S)$ such that its fixed points are equivalent to zeros of this gradient expression in terms of $\Delta_S = \widetilde{\Sigma}_{e,S} - \Sigma_{e,S}$. Then we show that the function F maps the ℓ_∞-ball $\mathbb{B}(L)$ onto itself, where

$$\mathbb{B}(L) := \{\Delta_S : ||\Delta_S||_{\max} \leq L\}, \tag{11.18}$$

with $L = 2\kappa_{\Gamma^*}(||W||_{\max} + \lambda)$. With these results, we can apply Brouwer's fixed point theorem to obtain that the mapping F has a fixed point inside the ball $\mathbb{B}(L)$.

For any matrix A, X, B, the derivatives of traces have the following expressions

$$\frac{\partial Tr(AX^{-1}B)}{\partial X} = -(X^{-1}BAX)^\top, \quad \frac{\partial Tr(X^{-1}B)}{\partial X} = -(X^{-1}BX^{-1})^\top.$$

The zero-gradient condition is

$$G\left(\widetilde{\Sigma}_e\right) = \left(\widetilde{\Sigma}_e^{-1}\right)_S - \left(\widetilde{\Sigma}_e^{-1} S_e \widetilde{\Sigma}_e^{-1}\right)_S + \lambda_n \widetilde{Z}_S = 0, \tag{11.19}$$

Note that $vec(\Delta_S) = vec(\widetilde{\Sigma}_{e,S}) - vec(\Sigma_{e,S})$. Define

$$F(vec(\Delta_S)) := -\left(\Gamma_{SS}^*\right)^{-1} vec(G(\Sigma_{e,S} + \Delta_S)) + vec(\Delta_S).$$

So $F(vec(\Delta_S)) = vec(\Delta_S)$ if and only if $G(\Sigma_{e,S} + \Delta_S) = 0$. Let $\Delta \in \mathbb{R}^{p\times p}$ be the zero-padded matrix, which is equal to Δ_S on the set S and zero on S^c.

Now we decompose $G(\Sigma_{e,S} + \Delta_S)$ below

$$\begin{aligned} G(\Sigma_{e,S} + \Delta_S) &= (\Sigma_e + \Delta)_S^{-1} - \left(\widetilde{\Sigma}_e^{-1} S_e \widetilde{\Sigma}_e^{-1}\right)_S + \lambda_n \widetilde{Z}_S \\ &= (\Sigma_e + \Delta)_S^{-1} - \Sigma_{e,S}^{-1} + V_S + \lambda_n \widetilde{Z}_S, \end{aligned} \tag{11.20}$$

where $V_S := \Sigma_{e,S}^{-1} - \left(\widetilde{\Sigma}_e^{-1} S_e \widetilde{\Sigma}_e^{-1}\right)_S$.

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For $F(vec(\Delta_S))$, we have the following evaluation, in terms of the decomposition (11.20),

$$\begin{aligned}
F(vec(\Delta_S)) &= -\left(\Gamma^*_{SS}\right)^{-1} \cdot vec(G(\Sigma_{e,S} + \Delta_S)) - vec(\Delta_S) \\
&= \left(\Gamma^*_{SS}\right)^{-1} \cdot vec\left[\Sigma^{-1}_{e,S} - (\Sigma_e + \Delta)^{-1}_S - V_S - \lambda \widetilde{Z}_S\right] - vec(\Delta_S) \\
&= \left(\Gamma^*_{SS}\right)^{-1} \cdot \left[\Gamma^*_{SS} vec(\Delta_S) - vec\left((\Sigma_e^{-1}\Delta)^2 J \Sigma_e^{-1}\right)_S\right] - \left(\Gamma^*_{SS}\right)^{-1} (V_S + \lambda \widetilde{Z}_S) - vec(\Delta_S) \\
&= -\left(\Gamma^*_{SS}\right)^{-1} \cdot vec\left((\Sigma_e^{-1}\Delta)^2 J \Sigma_e^{-1}\right)_S - \left(\Gamma^*_{SS}\right)^{-1} (V_S + \lambda \widetilde{Z}_S).
\end{aligned}$$

The two terms above can be evaluated as follows.

$$\left|\left|\left(\Gamma^*_{SS}\right)^{-1} \left(V_S + \lambda \widetilde{Z}_S\right)\right|\right| \leq \kappa_{\Gamma^*} \left(||V_S|| + \lambda\right) = \frac{L}{2},$$

and

$$\begin{aligned}
\left|\left|\left(\Gamma^*_{SS}\right)^{-1} vec\left((\Sigma_e^{-1}\Delta)^2 J \Sigma_e^{-1}\right)_S\right|\right|_\infty &\leq \kappa_{\Gamma^*} \cdot ||R(\Delta)||_{\max} \leq \kappa_{\Gamma^*} \cdot \frac{3}{2} d \cdot ||\Delta||^2_{\max} \cdot \kappa^3_{\Sigma_\ell} \\
&\leq \frac{3}{2} d \kappa_{\Gamma^*} \kappa^3_{\Sigma_\ell} \frac{1}{3\kappa^3_{\Sigma_e} \kappa_{\Gamma^*} d} = \frac{L}{2}.
\end{aligned}$$

□

Lemma 6 (Sign Consistency of $\widetilde{\Sigma}_e$)**.** *Suppose that conditions of Lemma 5 hold, and further that the minimum absolute value $\sigma_{\min}$ of nonzero entries in the true concentration matrix Σ_e is lower bounded as*

$$\sigma_{\min} \geq 4\kappa_{\Gamma^*} \left(||W||_{\max} + \lambda\right). \tag{11.21}$$

Then we have

$$sign\left(\widetilde{\Sigma}_{e,S}\right) = sign(\Sigma_{e,S}). \tag{11.22}$$

Proof. It follows from Lemma 5, for any $(i,j) \in S$,

$$\left|\widetilde{\Sigma}_{e,ij} - \Sigma_{e,ij}\right| \leq L.$$

By the bound of L, i.e.,

$$L \leq \min\{\frac{1}{3\kappa_{\Sigma_e} d}, \frac{1}{3\kappa^3_{\Sigma_e} \kappa_\Gamma d}\},$$

and the bound (11.21) on $\sigma_{\min}$, we could yield (11.22), i.e., the sign of $\widetilde{\Sigma}_{e,ij}$ can not differ from the sign of $\Sigma_{e,ij}$. □

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